

# The Physical Layer

## Chapter 2

# Theoretical Basis for Data Communication

- Fourier analysis
- Bandwidth-limited signals
- Maximum data rate of a channel

# Fourier Analysis

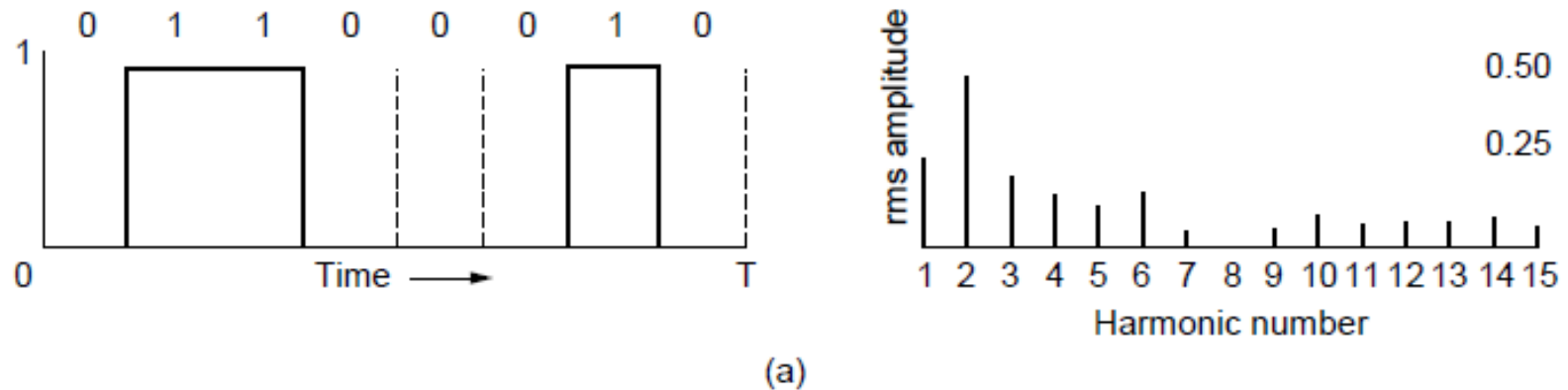
- We model the behavior of variation of voltage or current with mathematical functions
- Fourier series is used

$$g(t) = \frac{1}{2}c + \sum_{n=1}^{\infty} a_n \sin(2\pi nft) + \sum_{n=1}^{\infty} b_n \cos(2\pi nft)$$

- Function reconstructed with

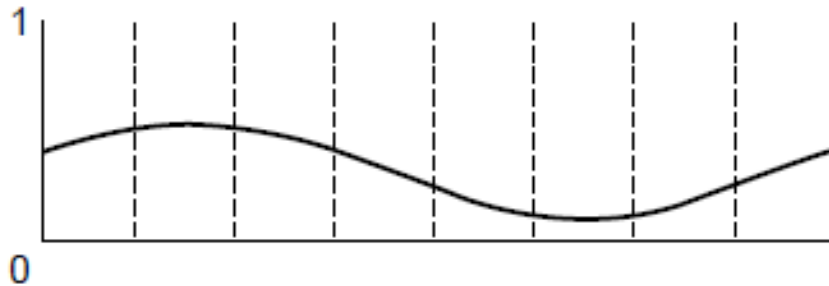
$$a_n = \frac{2}{T} \int_0^T g(t) \sin(2\pi nft) dt \quad b_n = \frac{2}{T} \int_0^T g(t) \cos(2\pi nft) dt \quad c = \frac{2}{T} \int_0^T g(t) dt$$

# Bandwidth-Limited Signals (1)

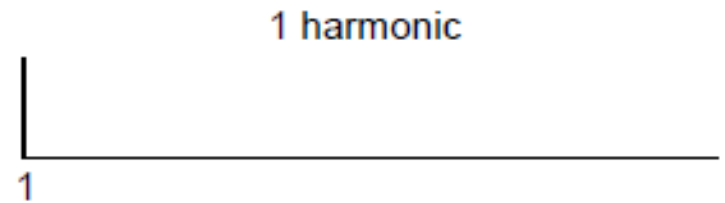


A binary signal and its root-mean-square  
Fourier amplitudes.

# Bandwidth-Limited Signals (2)

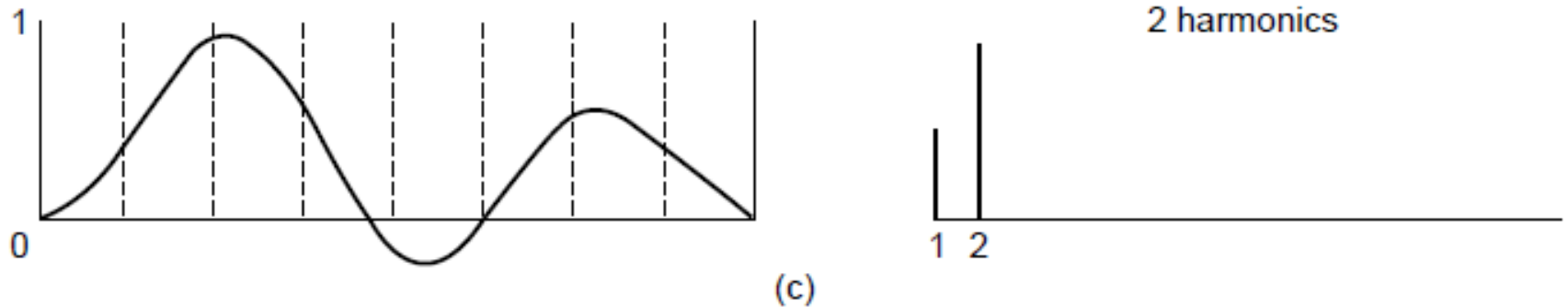


(b)



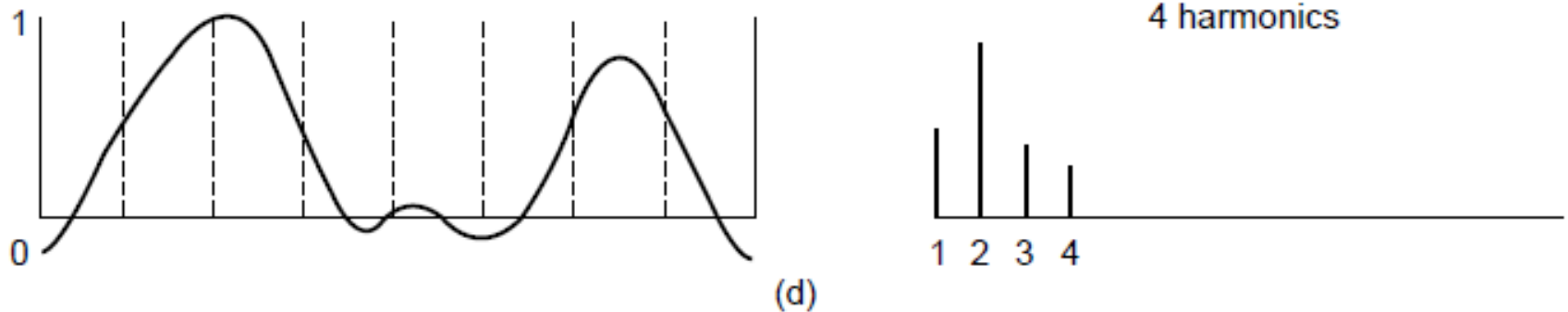
(b)-(e) Successive approximations to the original signal.

# Bandwidth-Limited Signals (3)



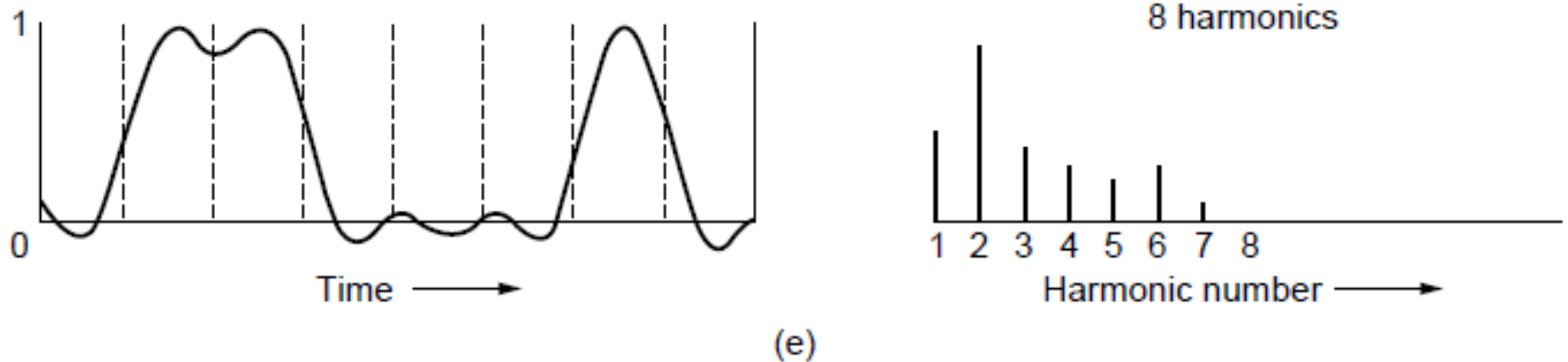
(b)-(e) Successive approximations to the original signal.

# Bandwidth-Limited Signals (4)



(b)-(e) Successive approximations to the original signal.

# Bandwidth-Limited Signals (5)



(b)-(e) Successive approximations to the original signal.

# Bandwidth-Limited Signals (6)

| <b>Bps</b> | <b>T (msec)</b> | <b>First harmonic (Hz)</b> | <b># Harmonics sent</b> |
|------------|-----------------|----------------------------|-------------------------|
| 300        | 26.67           | 37.5                       | 80                      |
| 600        | 13.33           | 75                         | 40                      |
| 1200       | 6.67            | 150                        | 20                      |
| 2400       | 3.33            | 300                        | 10                      |
| 4800       | 1.67            | 600                        | 5                       |
| 9600       | 0.83            | 1200                       | 2                       |
| 19200      | 0.42            | 2400                       | 1                       |
| 38400      | 0.21            | 4800                       | 0                       |

Relation between data rate and  
harmonics for our example.

# The Maximum Data Rate of a Channel

- Nyquist's theorem  
maximum data rate =  $2 B \log_2 \text{ bits / sec}$
- Shannon's formula for capacity of a noisy channel  
maximum number of bits / sec =  $B \log_2 (1 + S / N)$

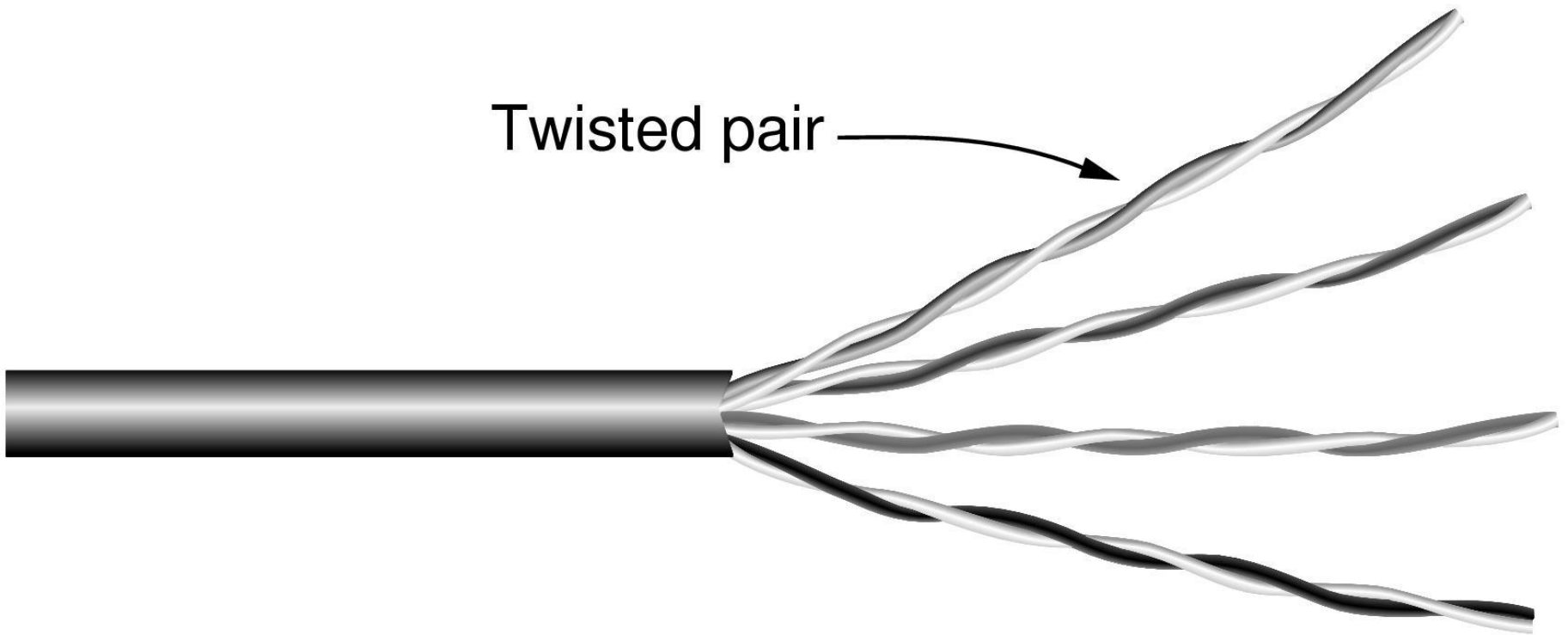
# Guided Transmission Media

- Magnetic media
- Twisted pairs
- Coaxial cable
- Power lines
- Fiber optics

# Magnetic Media

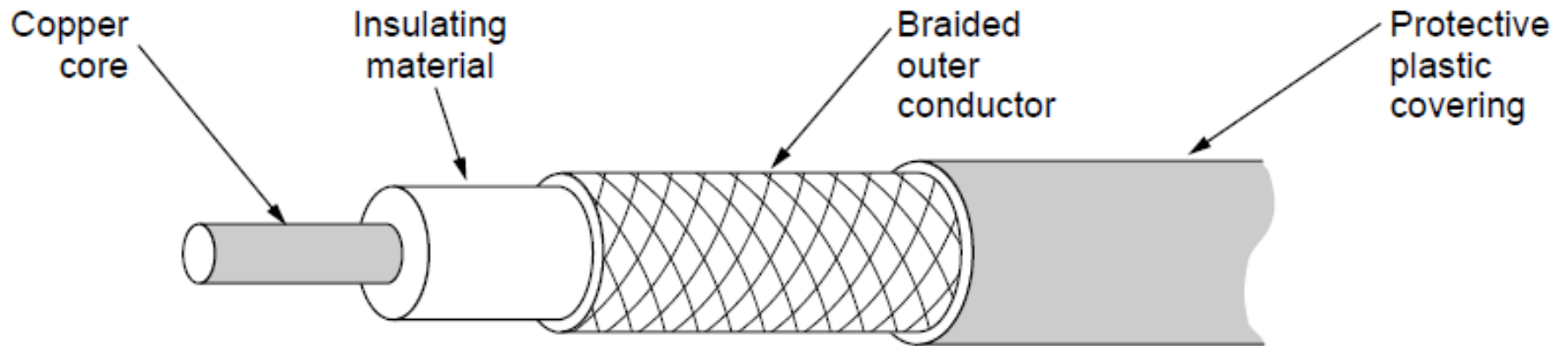
- Write data onto magnetic media
  - Disks
  - Tapes
- Data transmission speed
  - Never underestimate the bandwidth of a station wagon full of tapes hurtling down the highway.

# Twisted Pairs



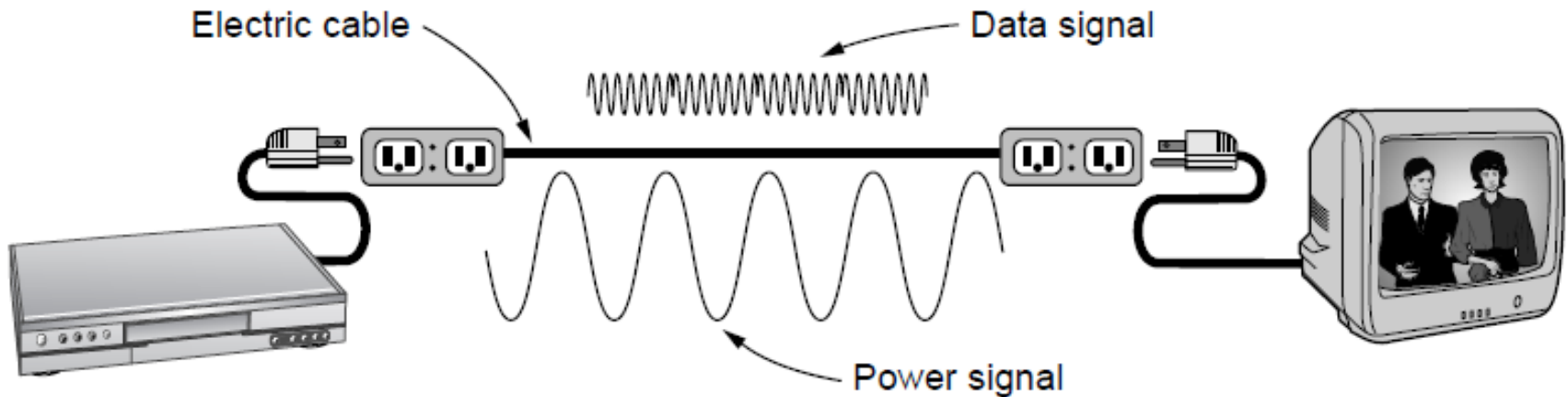
Category 5 UTP cable with four twisted pairs

# Coaxial Cable



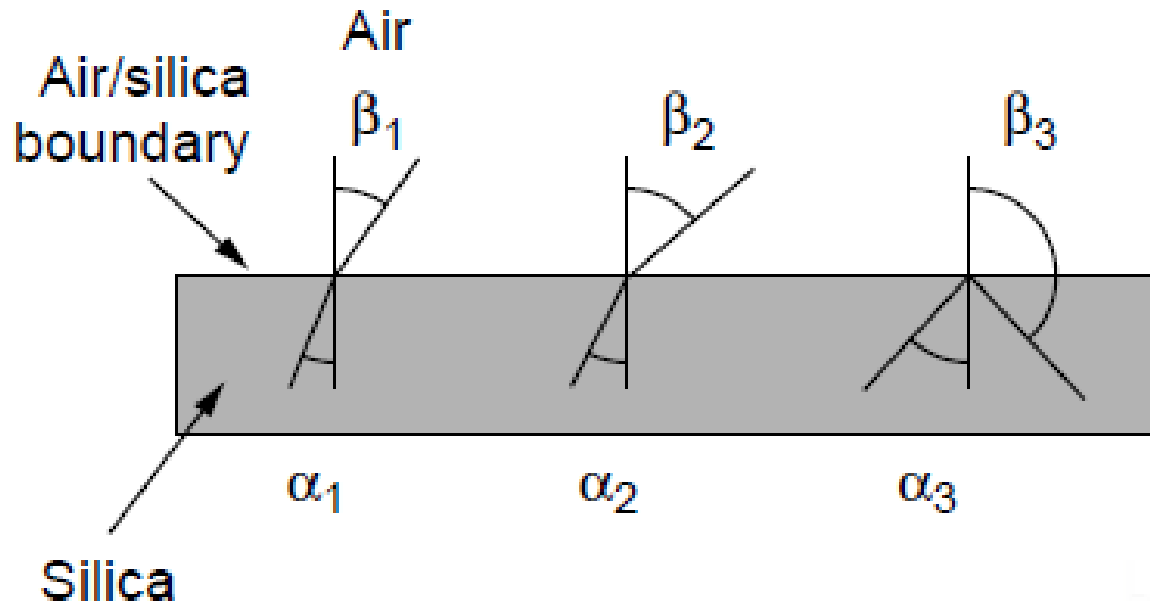
A coaxial cable

# Power Lines



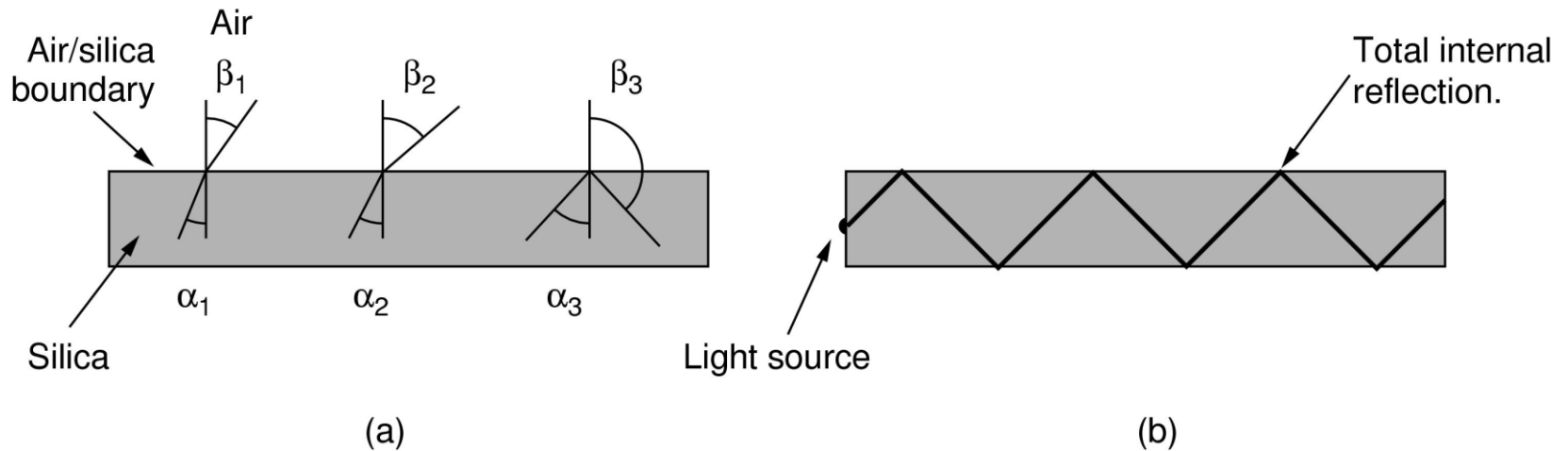
A network that uses household electrical wiring.

# Fiber Optics (1)



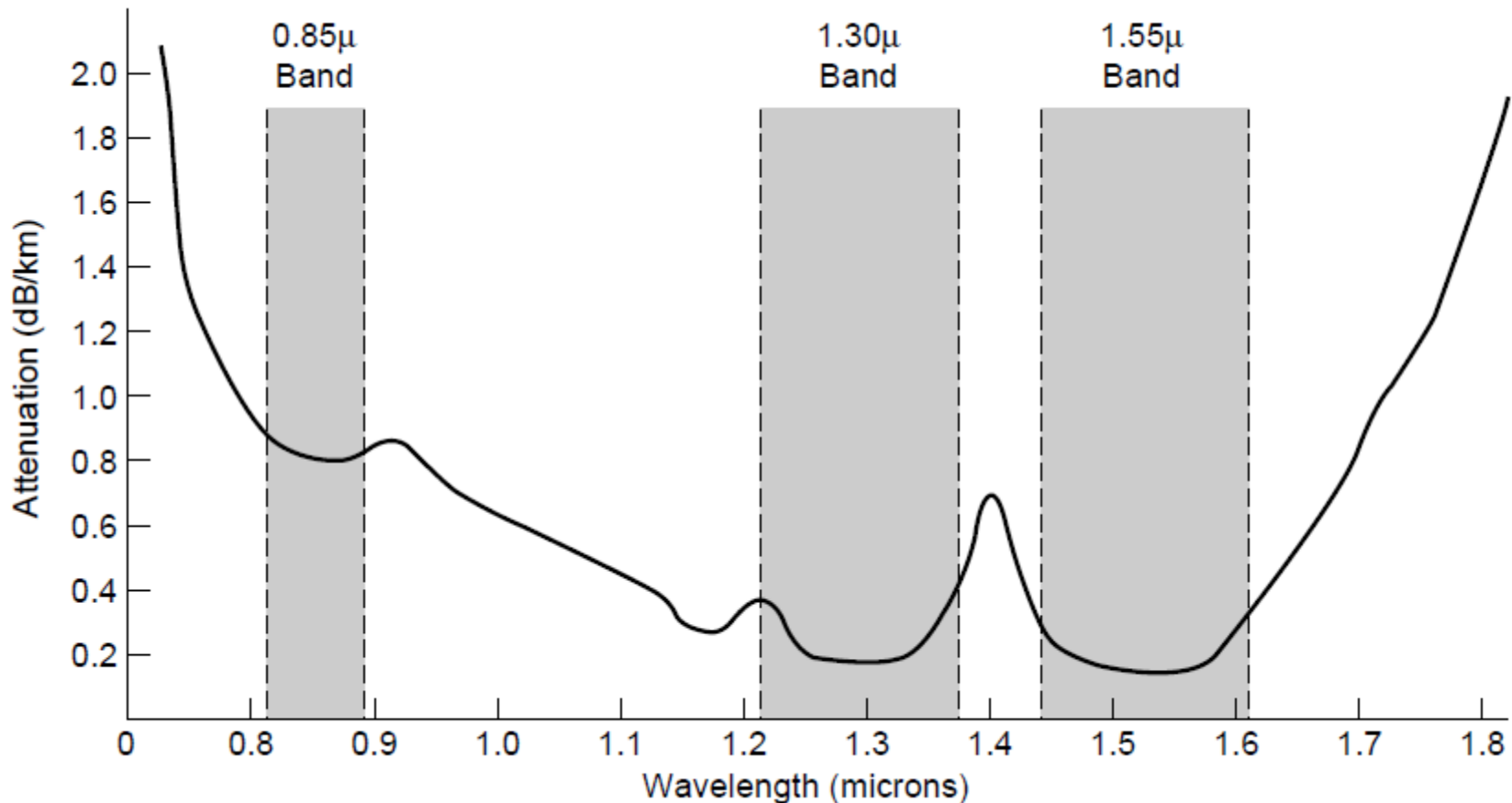
Three examples of a light ray from inside a silica fiber impinging on the air/silica boundary at different angles.

# Fiber Optics (2)



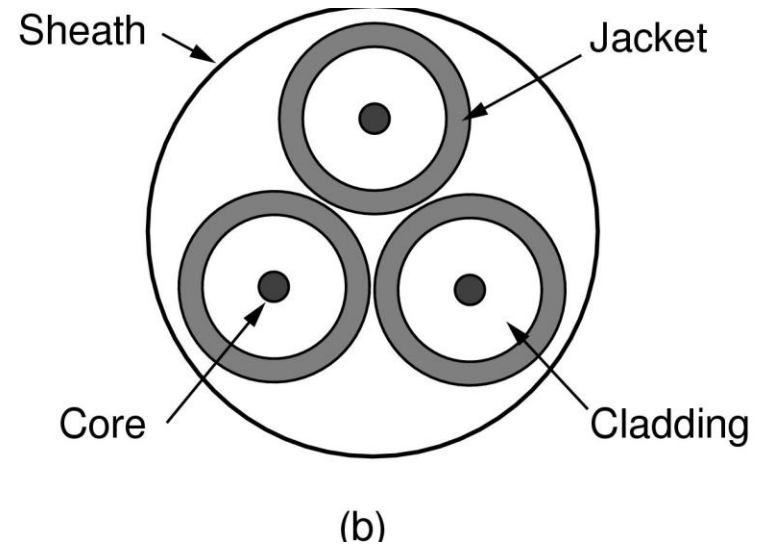
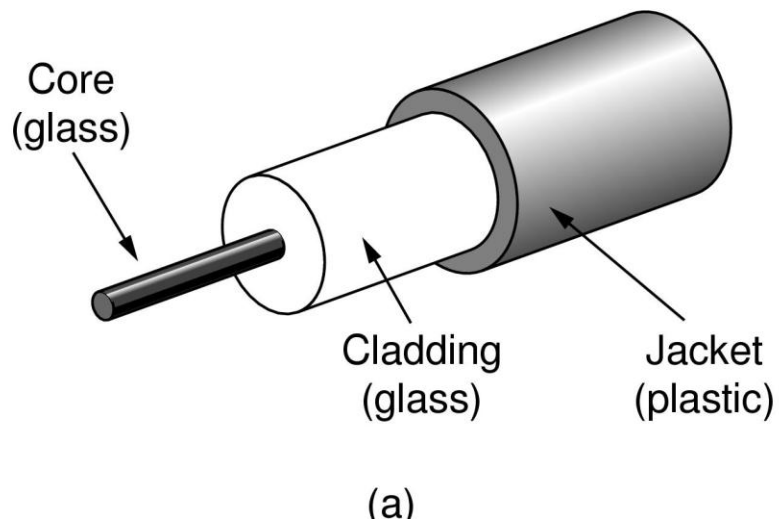
Light trapped by total internal reflection.

# Transmission of Light Through Fiber



Attenuation of light through fiber  
in the infrared region

# Fiber Cables (1)



Views of a fiber cable

## Fiber Cables (2)

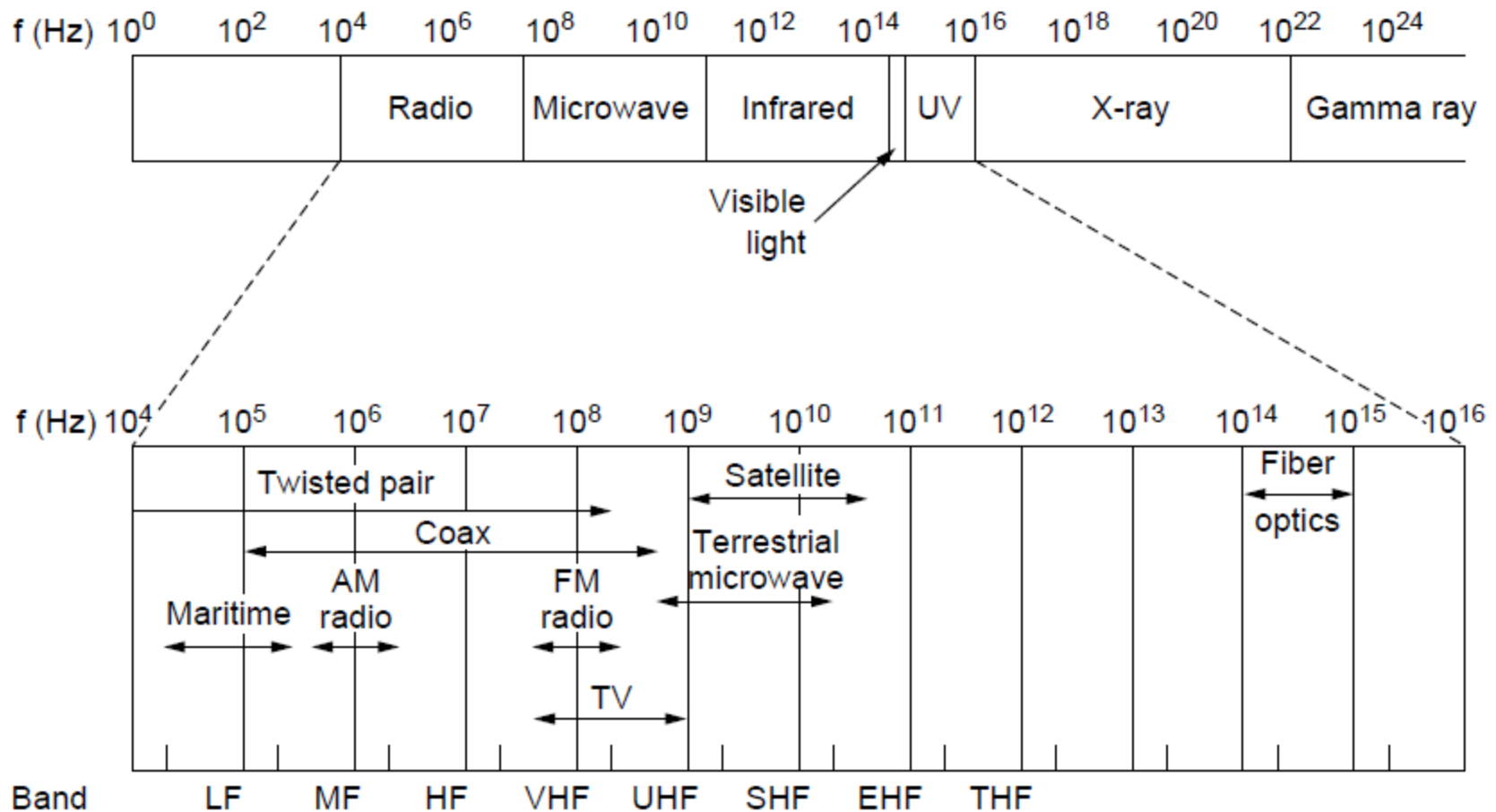
| Item                    | LED        | Semiconductor laser       |
|-------------------------|------------|---------------------------|
| Data rate               | Low        | High                      |
| Fiber type              | Multi-mode | Multi-mode or single-mode |
| Distance                | Short      | Long                      |
| Lifetime                | Long life  | Short life                |
| Temperature sensitivity | Minor      | Substantial               |
| Cost                    | Low cost   | Expensive                 |

A comparison of semiconductor diodes  
and LEDs as light sources

# Wireless Transmission

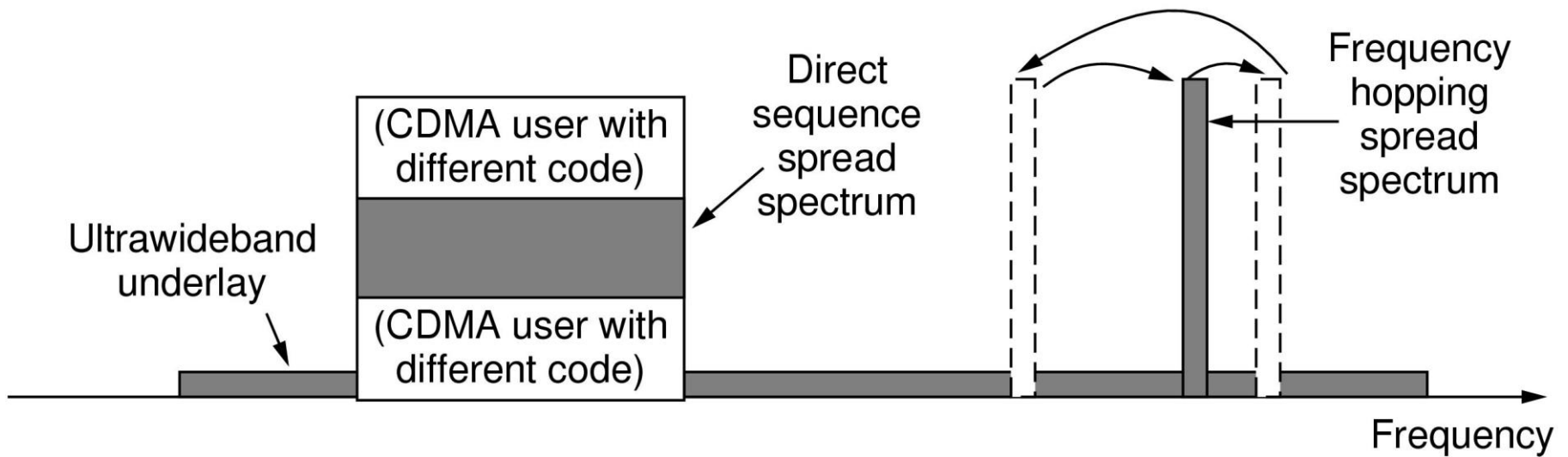
- The Electromagnetic Spectrum
- Radio Transmission
- Microwave Transmission
- Infrared Transmission
- Light Transmission

# The Electromagnetic Spectrum (1)



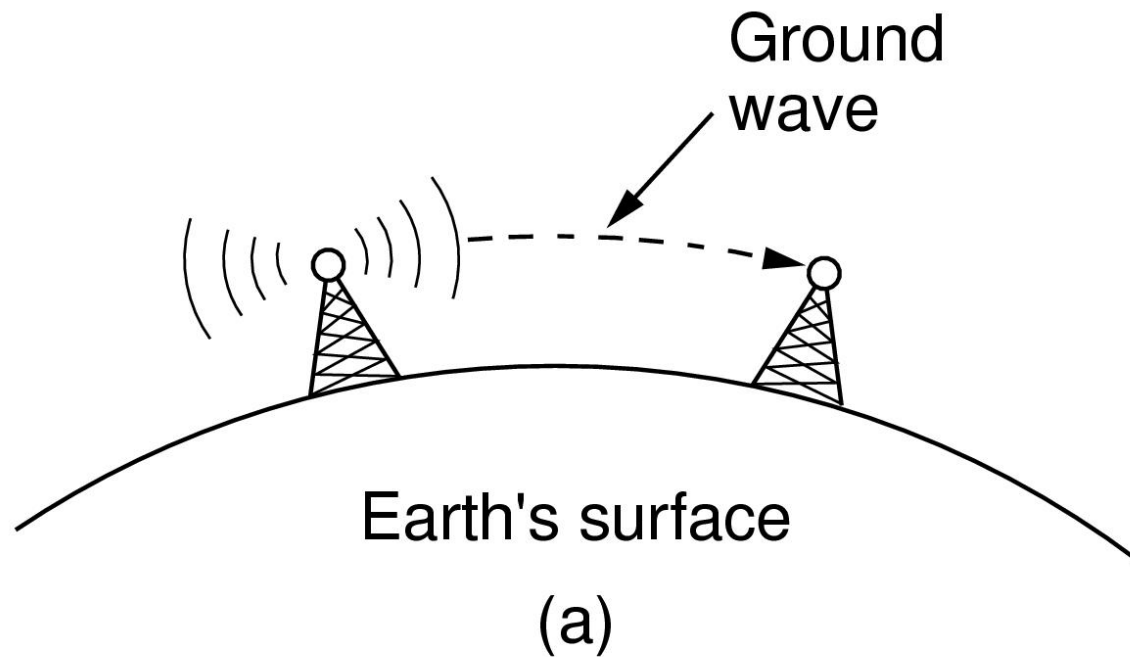
The electromagnetic spectrum and  
its uses for communication

# The Electromagnetic Spectrum (2)



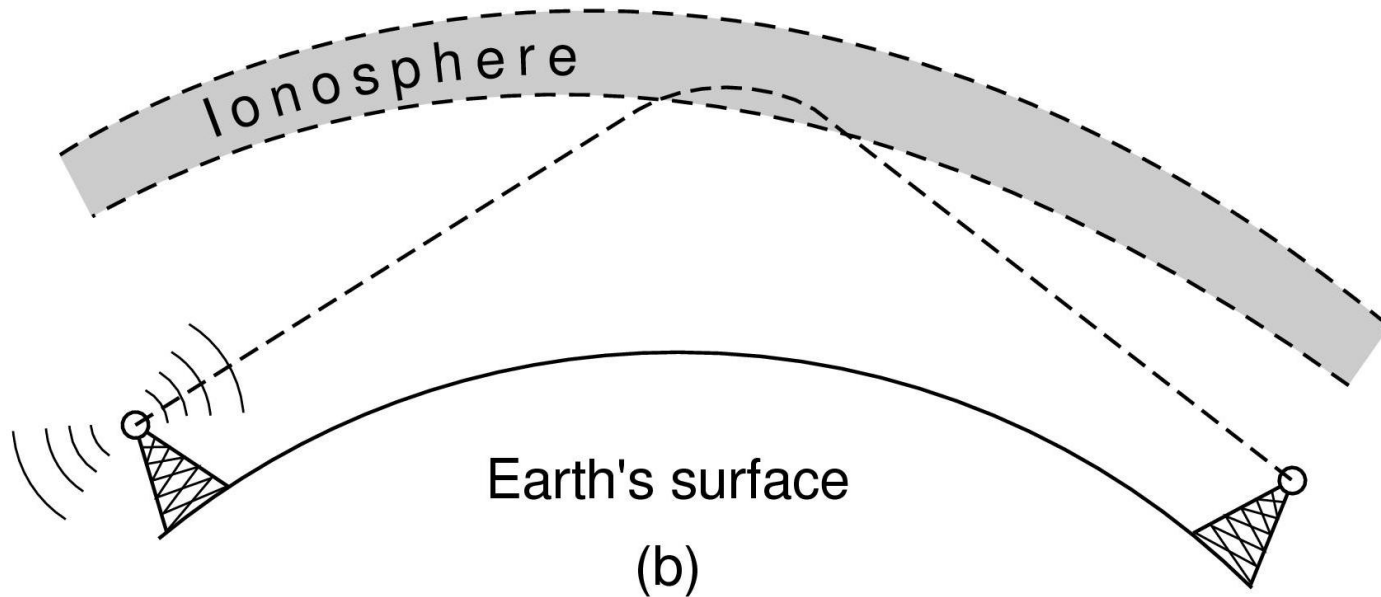
Spread spectrum and ultra-wideband  
(UWB) communication

# Radio Transmission (1)



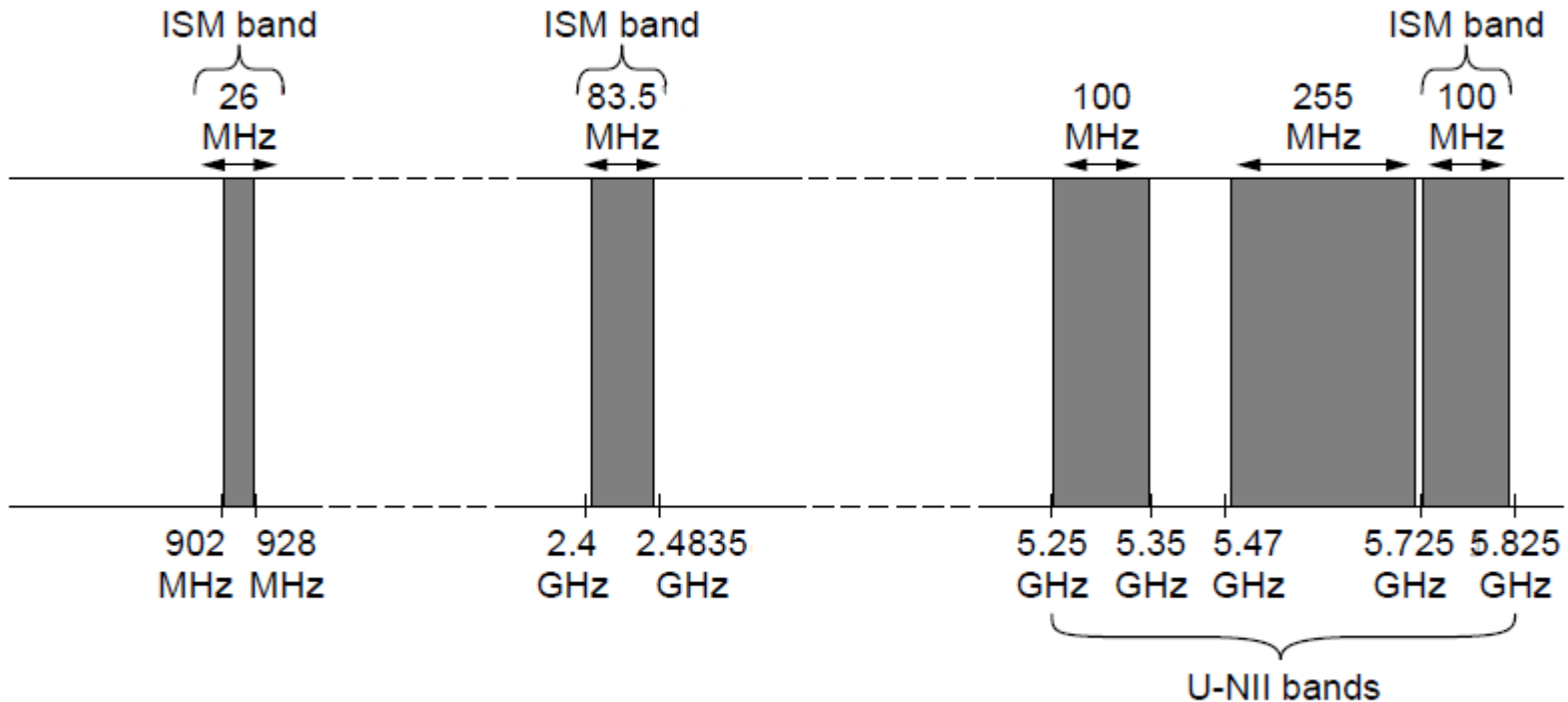
In the VLF, LF, and MF bands, radio waves follow the curvature of the earth

## Radio Transmission (2)



In the HF band, they bounce off the ionosphere.

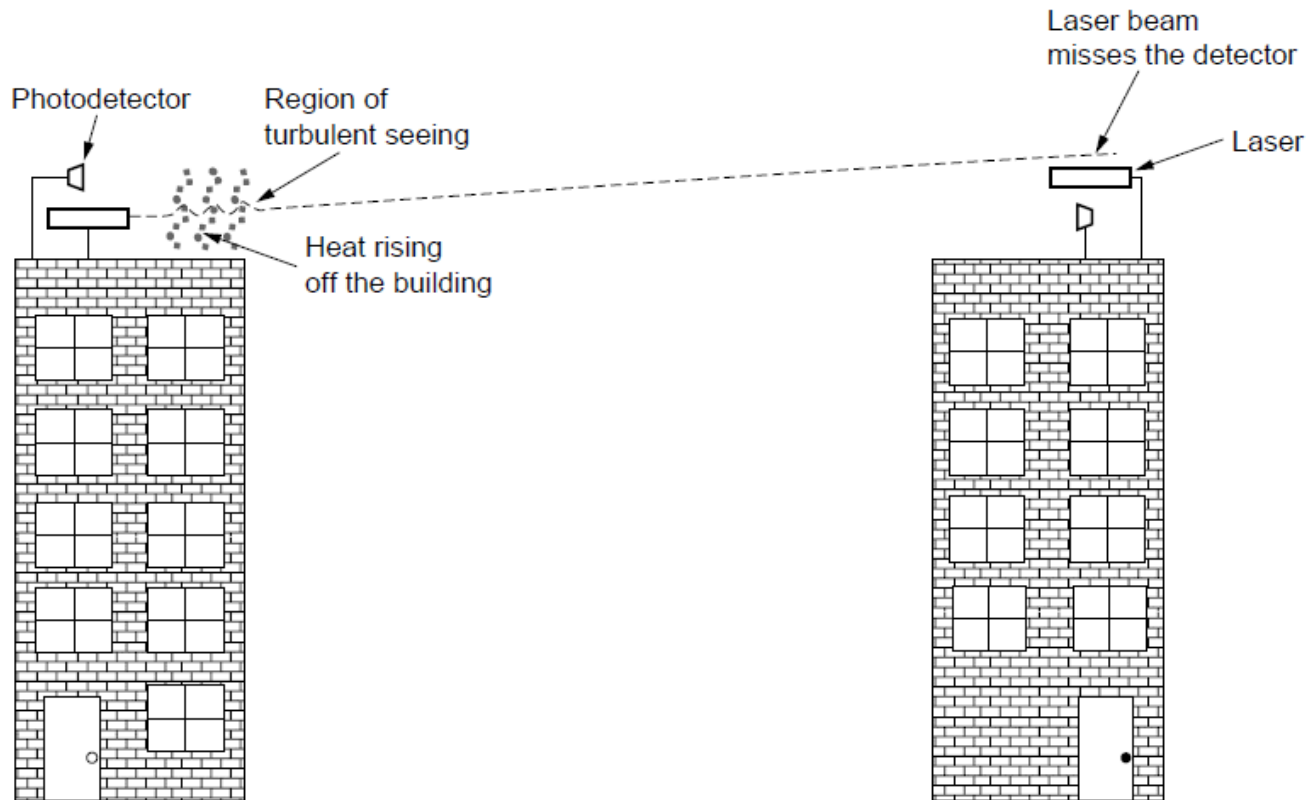
# The Politics of the Electromagnetic Spectrum



ISM and U-NII bands used in the  
United States by wireless devices



# Light Transmission

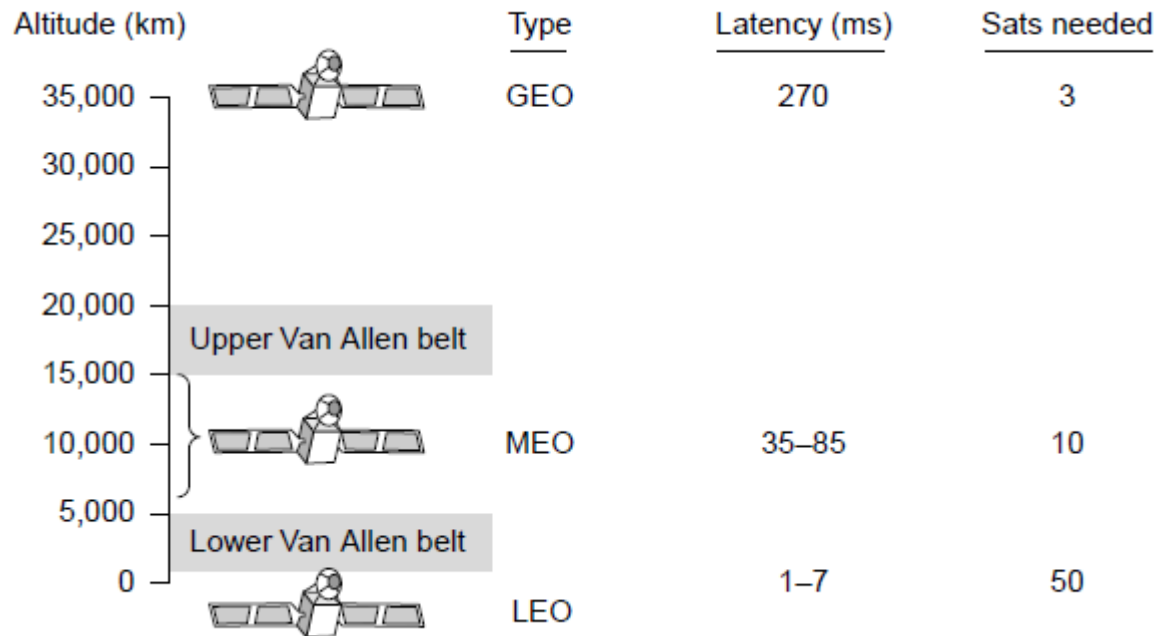


Convection currents can interfere with laser communication systems. A bidirectional system with two lasers is pictured here.

# Communication Satellites

- Geostationary Satellites
- Medium-Earth Orbit Satellites
- Low-Earth Orbit Satellites
- Satellites Versus Fiber

# Communication Satellites



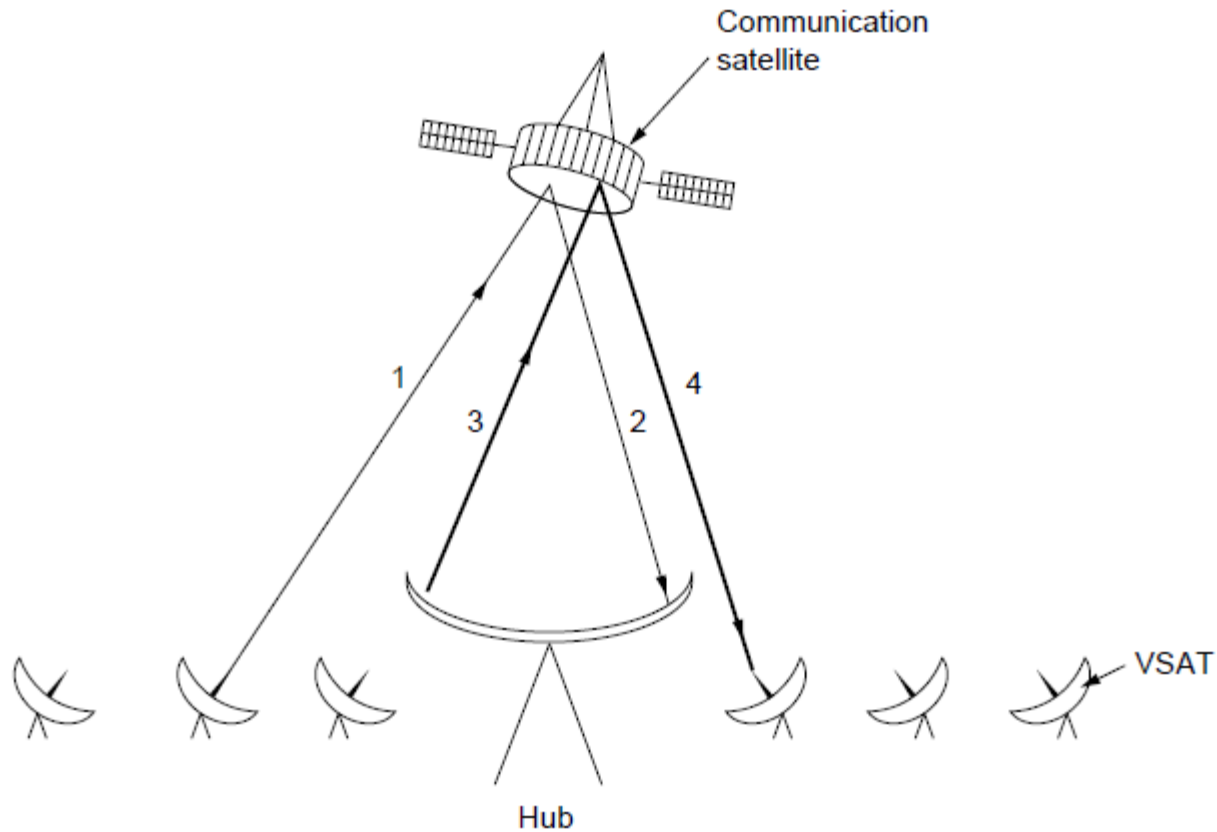
Communication satellites, some properties, including: altitude above earth, round-trip delay time, number of satellites for global coverage.

# Geostationary Satellites (1)

| Band | Downlink | Uplink  | Bandwidth | Problems                 |
|------|----------|---------|-----------|--------------------------|
| L    | 1.5 GHz  | 1.6 GHz | 15 MHz    | Low bandwidth; crowded   |
| S    | 1.9 GHz  | 2.2 GHz | 70 MHz    | Low bandwidth; crowded   |
| C    | 4.0 GHz  | 6.0 GHz | 500 MHz   | Terrestrial interference |
| Ku   | 11 GHz   | 14 GHz  | 500 MHz   | Rain                     |
| Ka   | 20 GHz   | 30 GHz  | 3500 MHz  | Rain, equipment cost     |

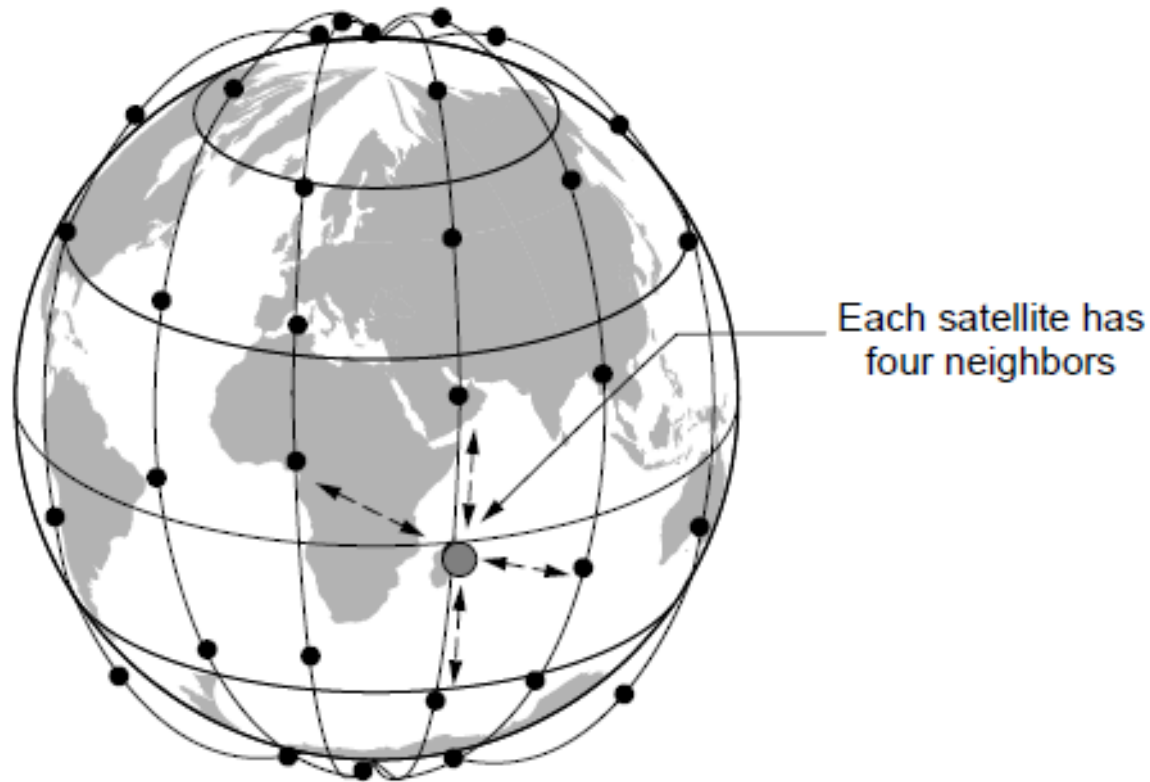
The principal satellite bands

# Geostationary Satellites (2)



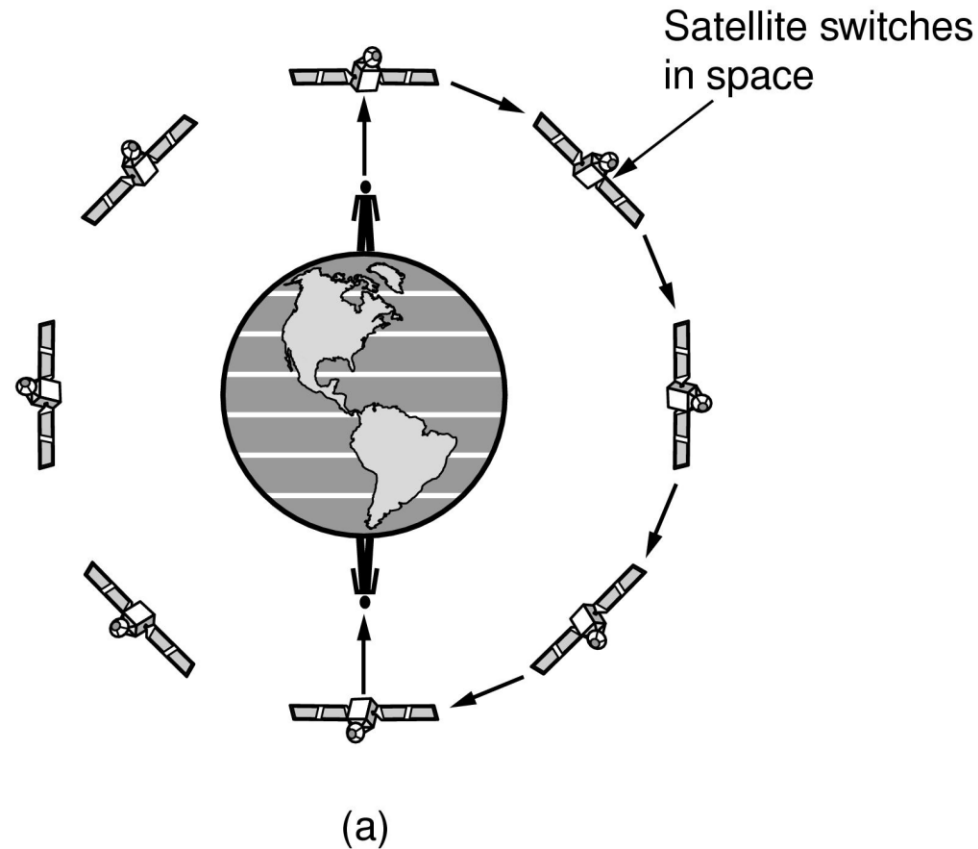
VSATs using a hub.

# Low-Earth Orbit Satellites (1)



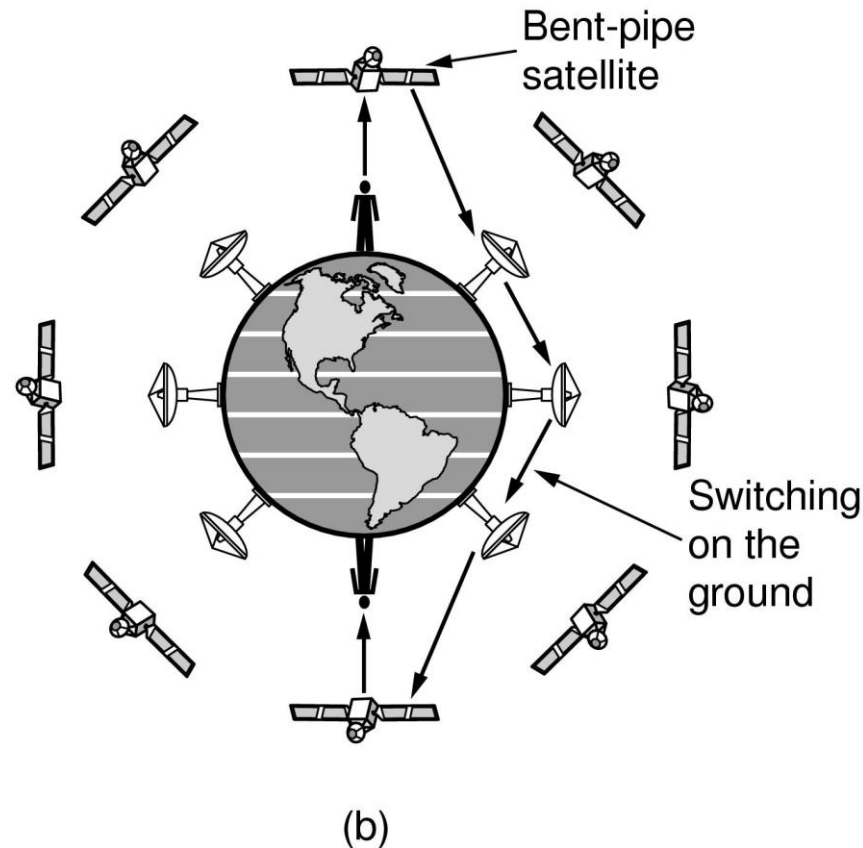
The Iridium satellites form six necklaces around the earth.

# Low-Earth Orbit Satellites (2)



Relaying in space.

# Low-Earth Orbit Satellites (3)

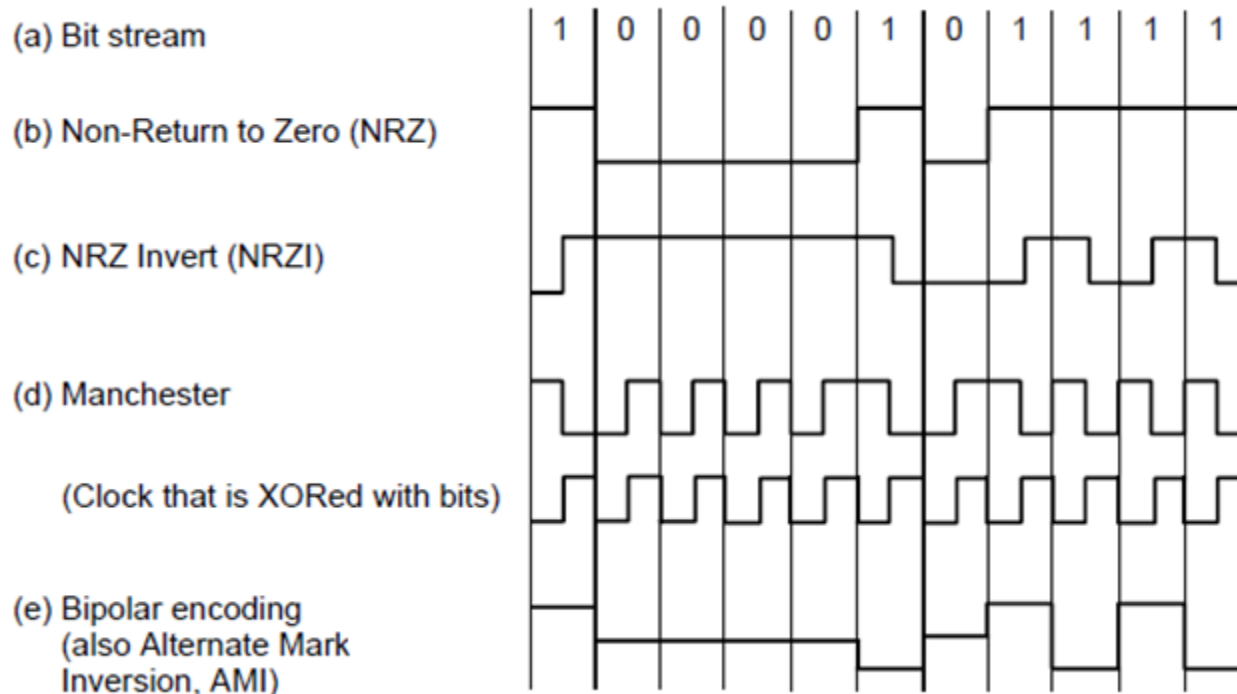


Relaying on the ground

# Digital Modulation and Multiplexing

- Baseband Transmission
- Passband Transmission
- Frequency Division Multiplexing
- Time Division Multiplexing
- Code Division Multiplexing

# Baseband Transmission



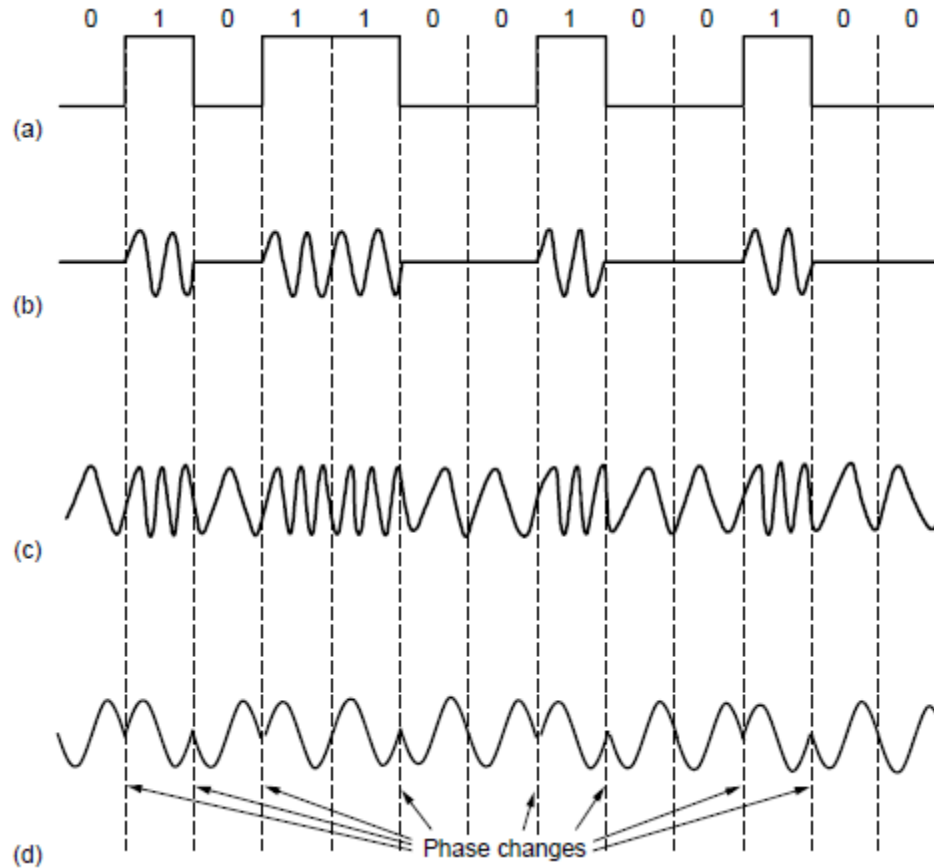
Line codes: (a) Bits, (b) NRZ, (c) NRZI,  
(d) Manchester, (e) Bipolar or AMI.

# Clock Recovery

| <b>Data (4B)</b> | <b>Codeword (5B)</b> | <b>Data (4B)</b> | <b>Codeword (5B)</b> |
|------------------|----------------------|------------------|----------------------|
| 0000             | 11110                | 1000             | 10010                |
| 0001             | 01001                | 1001             | 10011                |
| 0010             | 10100                | 1010             | 10110                |
| 0011             | 10101                | 1011             | 10111                |
| 0100             | 01010                | 1100             | 11010                |
| 0101             | 01011                | 1101             | 11011                |
| 0110             | 01110                | 1110             | 11100                |
| 0111             | 01111                | 1111             | 11101                |

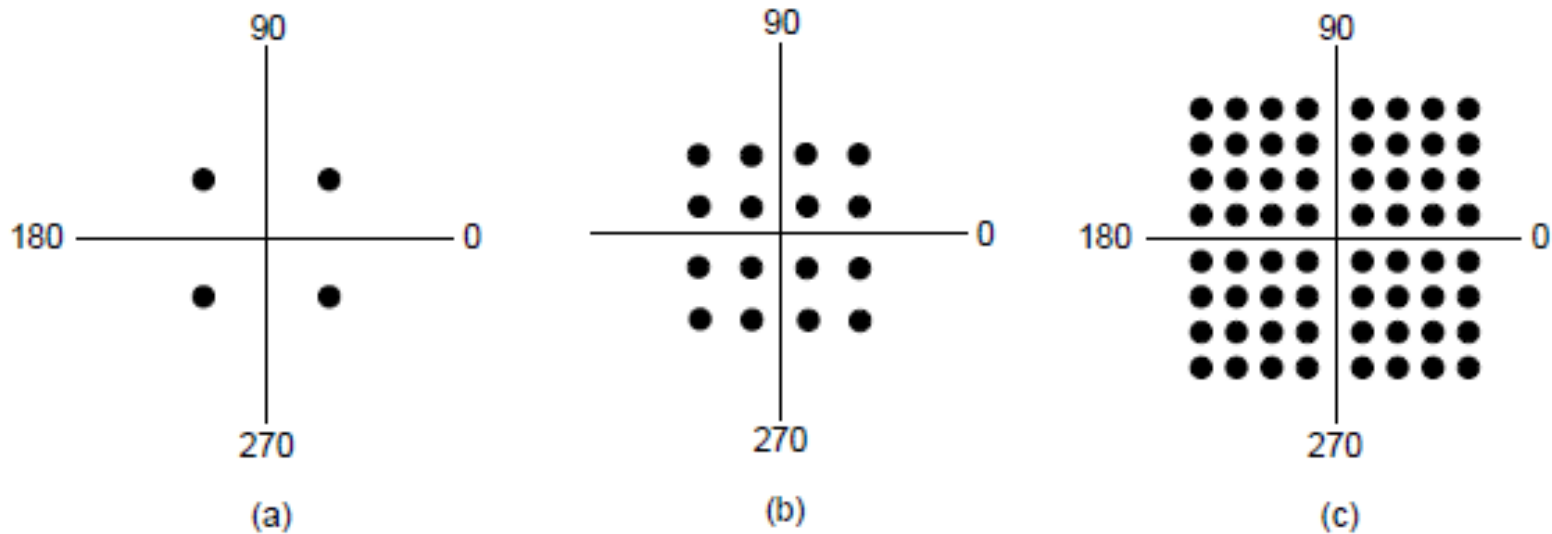
4B/5B mapping.

# Passband Transmission (1)



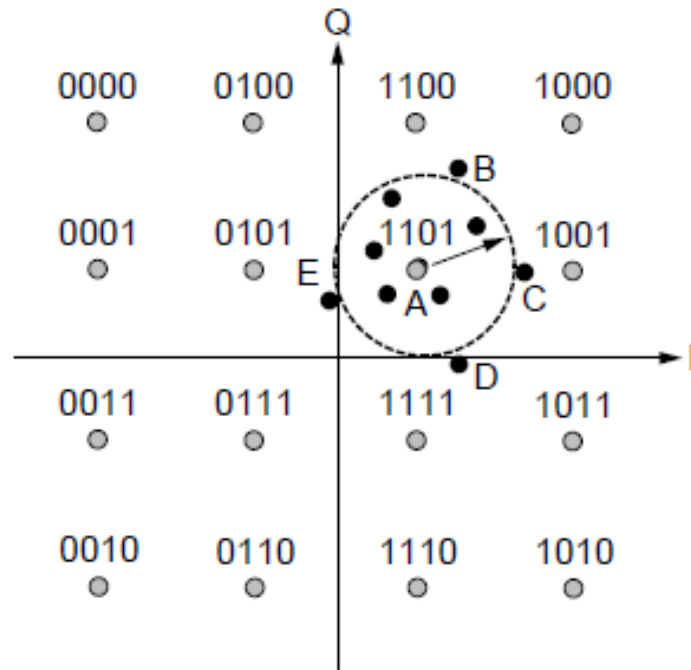
- (a) A binary signal. (b) Amplitude shift keying.  
(c) Frequency shift keying. (d) Phase shift keying.

# Passband Transmission (2)



(a) QPSK. (b) QAM-16. (c) QAM-64.

# Frequency Division Multiplexing (1)

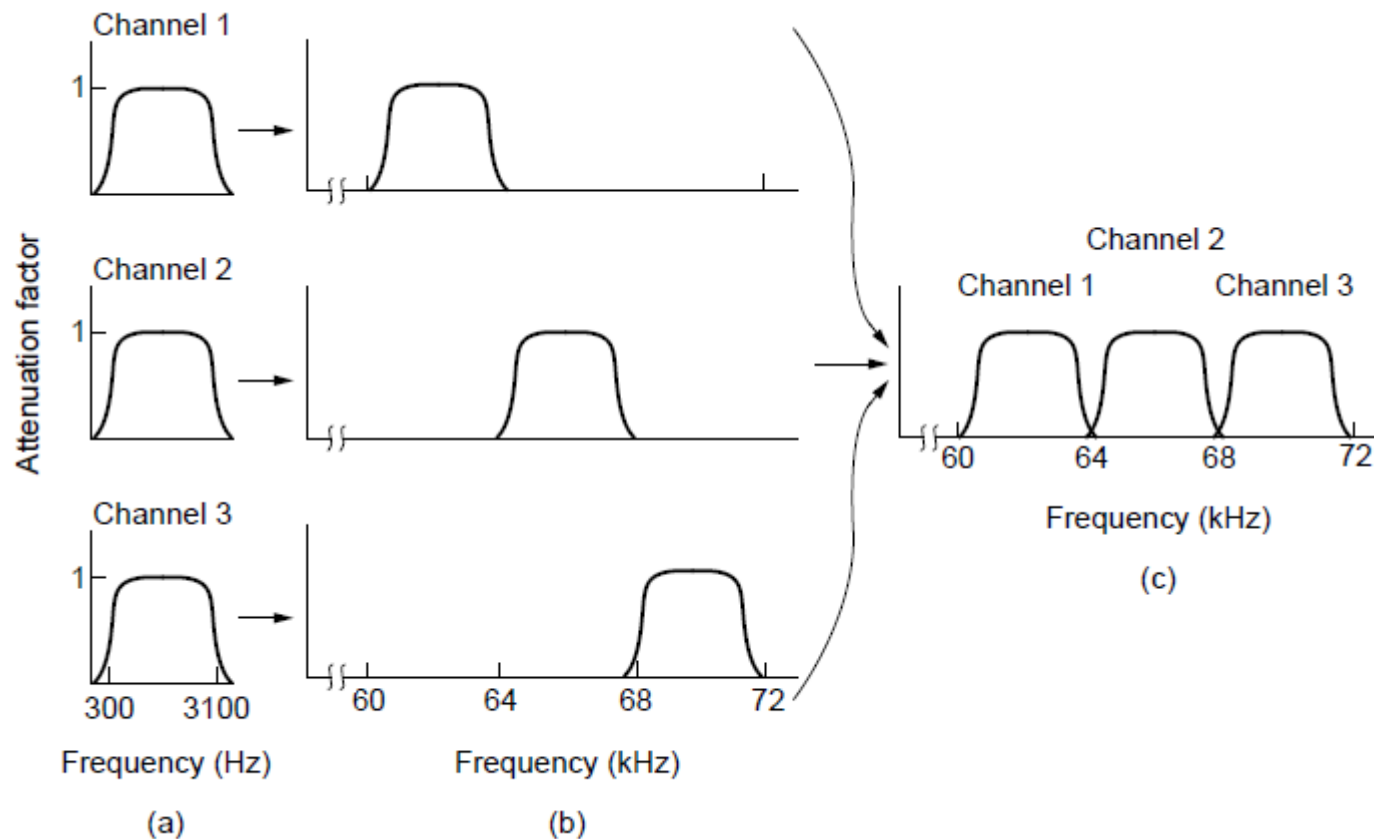


When 1101 is sent:

| Point | Decodes as    | Bit errors |
|-------|---------------|------------|
| A     | 1101          | 0          |
| B     | 110 <u>0</u>  | 1          |
| C     | <u>1</u> 001  | 1          |
| D     | 11 <u>1</u> 1 | 1          |
| E     | <u>0</u> 101  | 1          |

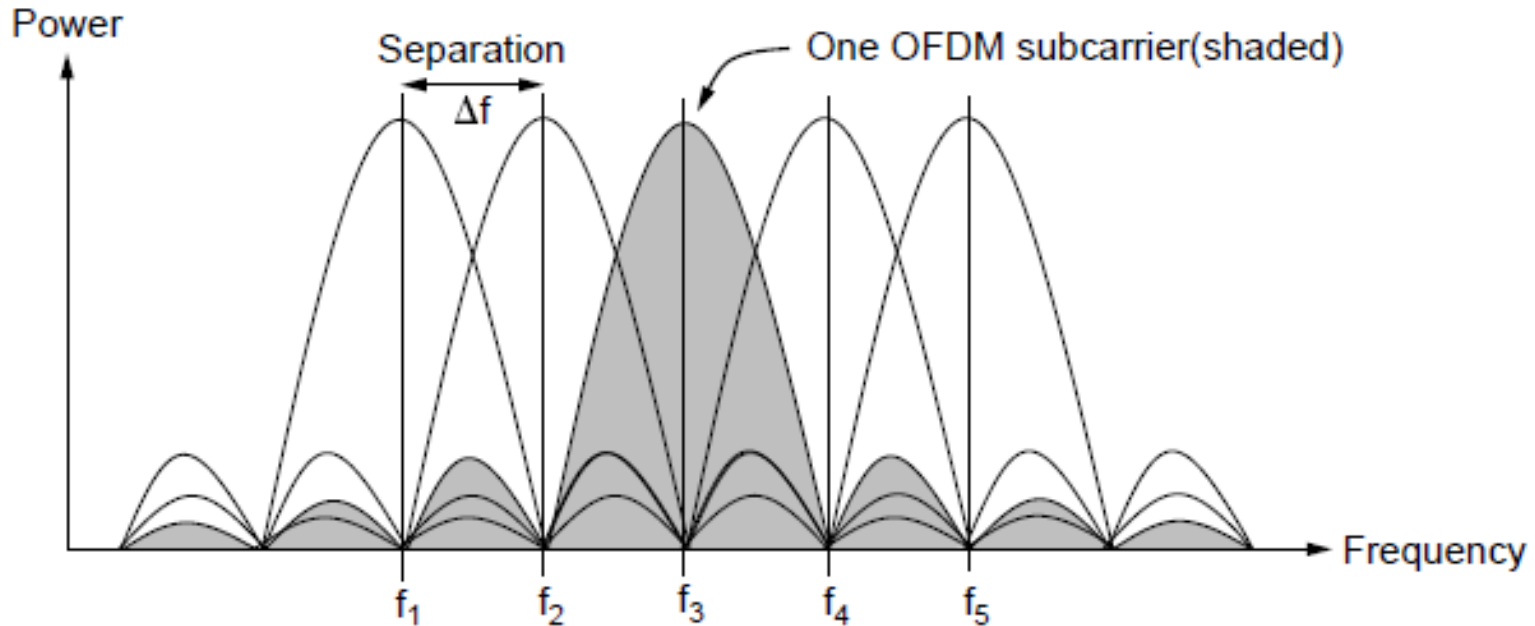
Gray-coded QAM-16.

# Frequency Division Multiplexing (2)



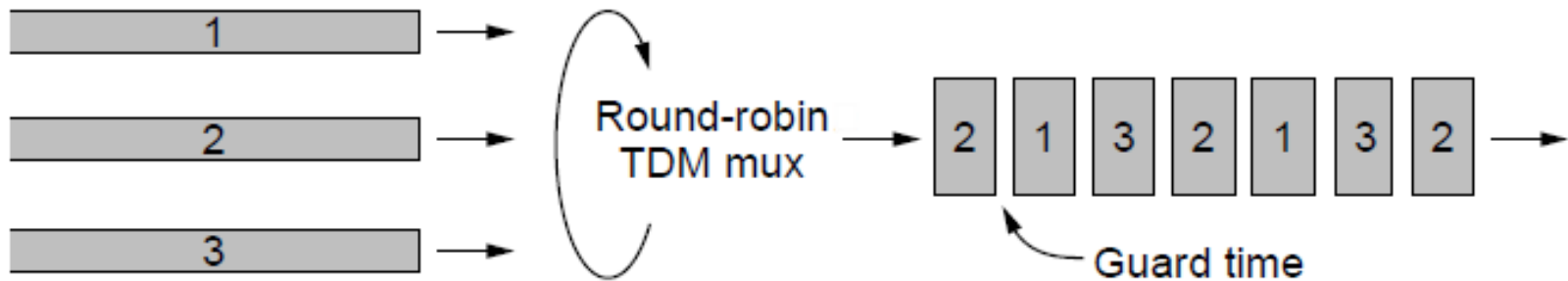
Frequency division multiplexing. (a) The original bandwidths.  
(b) The bandwidths raised in frequency.  
(c) The multiplexed channel.

# Frequency Division Multiplexing (3)



Orthogonal frequency division  
multiplexing (OFDM).

# Time Division Multiplexing



Time Division Multiplexing (TDM).

# Code Division Multiplexing (1)

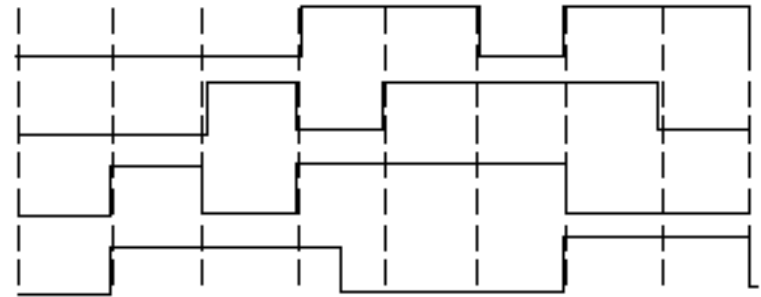
A = (-1 -1 -1 +1 +1 -1 +1 +1)

B = (-1 -1 +1 -1 +1 +1 +1 -1)

C = (-1 +1 -1 +1 +1 +1 -1 -1)

D = (-1 +1 -1 -1 -1 -1 +1 -1)

(a)



(b)

(a) Chip sequences for four stations.

(b) Signals the sequences represent

# Code Division Multiplexing (2)

|                                  |                                             |                                            |
|----------------------------------|---------------------------------------------|--------------------------------------------|
| $S_1 = C$                        | $= (-1 \ +1 \ -1 \ +1 \ +1 \ +1 \ -1 \ -1)$ | $S_1 \bullet C = [1+1-1+1+1+1-1-1]/8 = 1$  |
| $S_2 = B + \overline{C}$         | $= (-2 \ 0 \ 0 \ 0 \ +2 \ +2 \ 0 \ -2)$     | $S_2 \bullet C = [2+0+0+0+2+2+0+2]/8 = 1$  |
| $S_3 = A + \overline{B}$         | $= (0 \ 0 \ -2 \ +2 \ 0 \ -2 \ 0 \ +2)$     | $S_3 \bullet C = [0+0+2+2+0-2+0-2]/8 = 0$  |
| $S_4 = A + \overline{B} + C$     | $= (-1 \ +1 \ -3 \ +3 \ +1 \ -1 \ -1 \ +1)$ | $S_4 \bullet C = [1+1+3+3+1-1+1-1]/8 = 1$  |
| $S_5 = A + B + \overline{C} + D$ | $= (-4 \ 0 \ -2 \ 0 \ +2 \ 0 \ +2 \ -2)$    | $S_5 \bullet C = [4+0+2+0+2+0-2+2]/8 = 1$  |
| $S_6 = A + B + \overline{C} + D$ | $= (-2 \ -2 \ 0 \ -2 \ 0 \ -2 \ +4 \ 0)$    | $S_6 \bullet C = [2-2+0-2+0-2-4+0]/8 = -1$ |

(c)
(d)

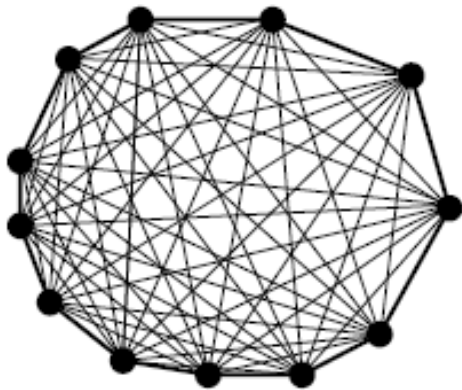
(a) Six examples of transmissions.

(b) Recovery of station C's

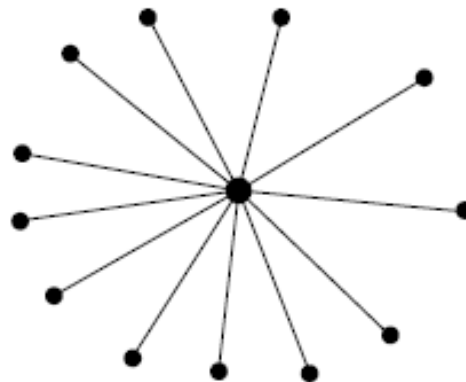
# The Public Switched Telephone Network

- Structure of the telephone system
- Politics of telephones
- Local loop: modems, ADSL, and fiber
- Trunks and multiplexing
- Switching

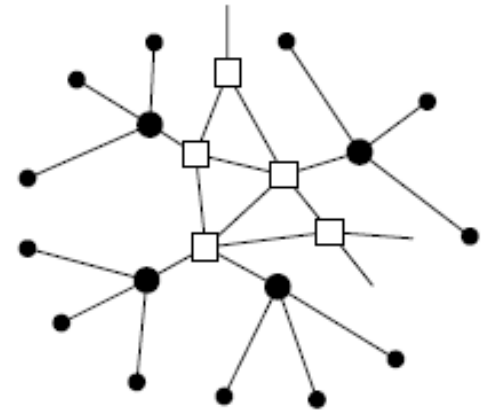
# Structure of the Telephone System (1)



(a)



(b)



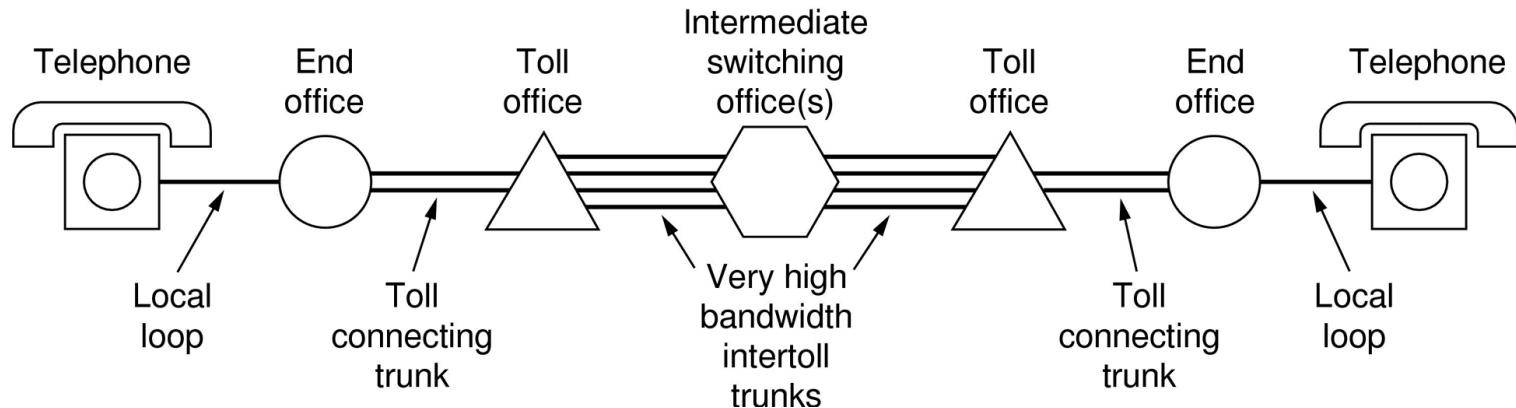
(c)

(a) Fully interconnected network.

(b) Centralized switch.

(c) Two-level hierarchy.

# Structure of the Telephone System (2)



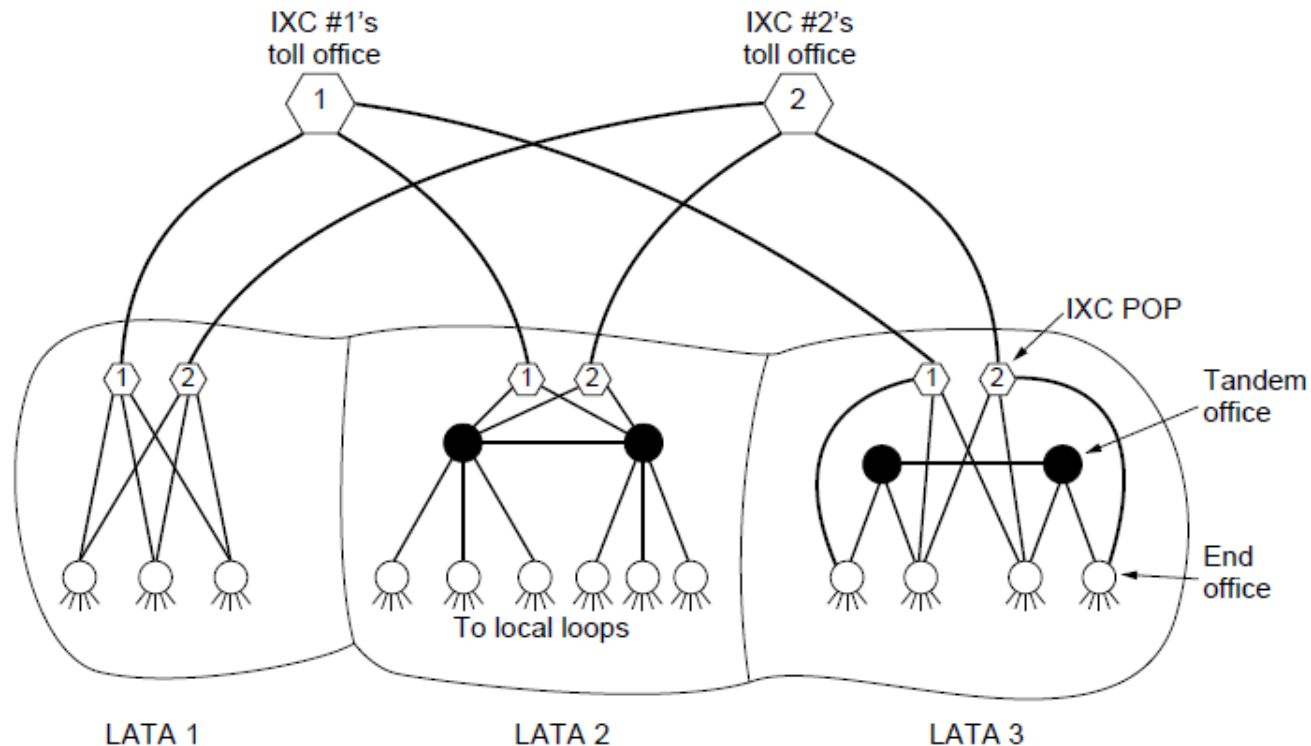
A typical circuit route for a long-distance call.

# Structure of the Telephone System (3)

## Major Components

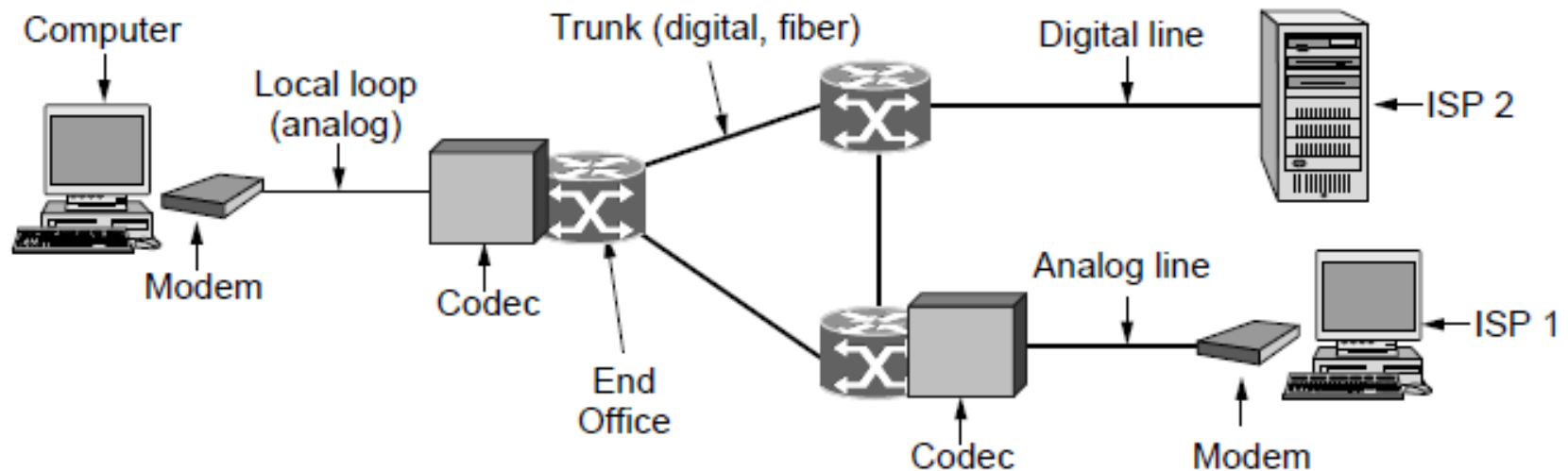
1. Local loops analog twisted pairs to houses, businesses).
2. Trunks (digital fiber optic links between switching offices).
3. Switching offices (calls are moved from one trunk to another).

# The Politics of Telephones



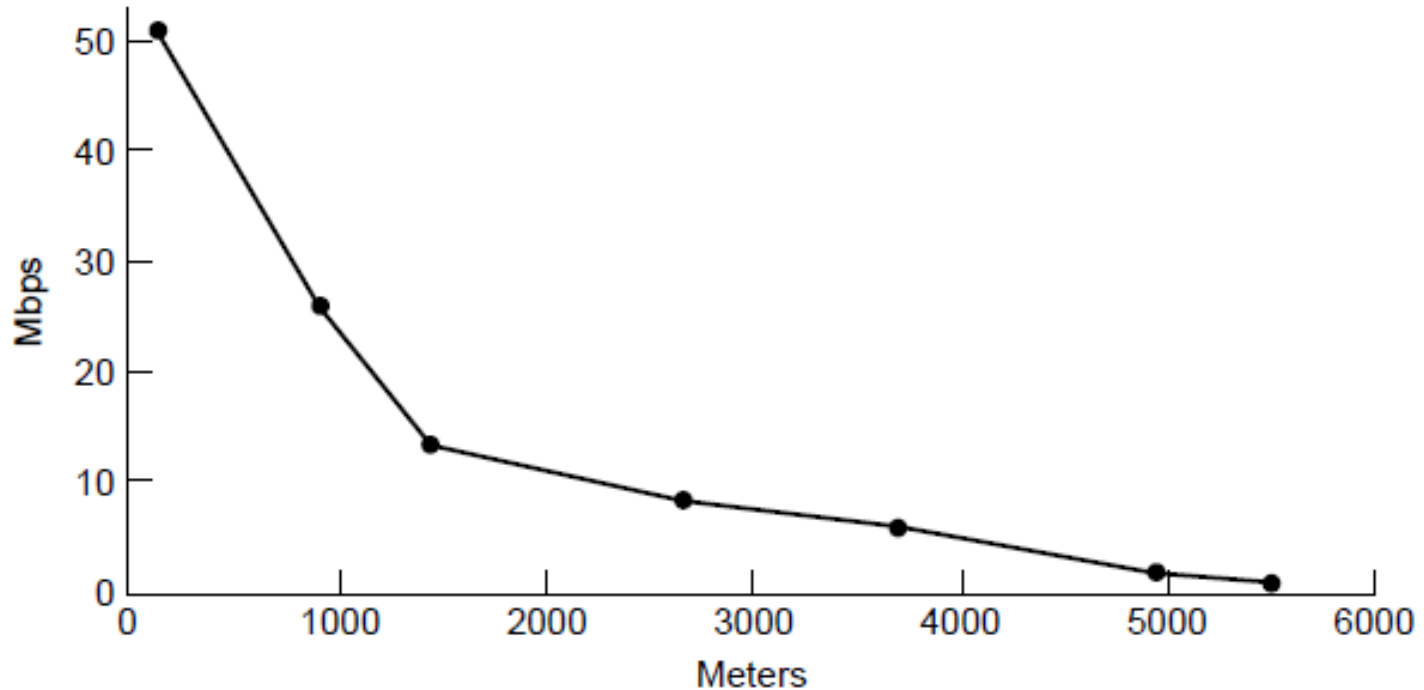
The relationship of LATAs, LECs, and IXCs. Circles are LEC switching offices. Hexagons belong to IXC whose number is in it.

# Telephone Modems



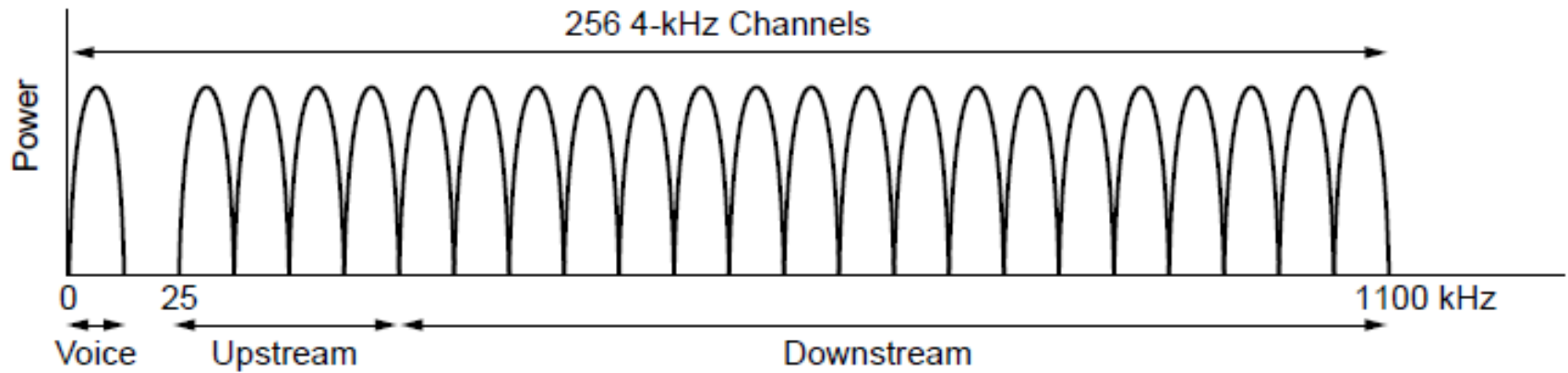
Use of both analog and digital transmission for computer -to-computer call. Conversion done by modems and codecs.

# Digital Subscriber Lines (1)



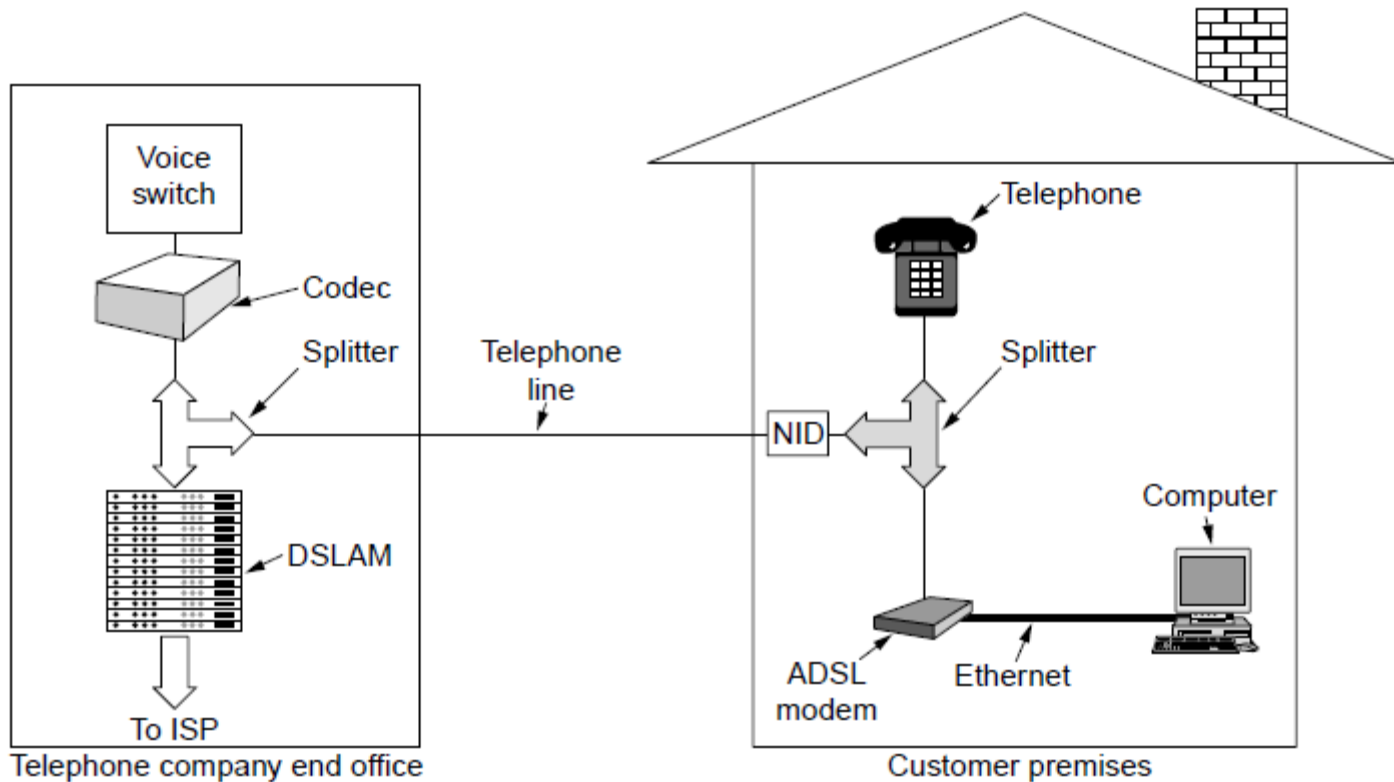
Bandwidth versus distance over Category 3  
UTP for DSL.

# Digital Subscriber Lines (2)



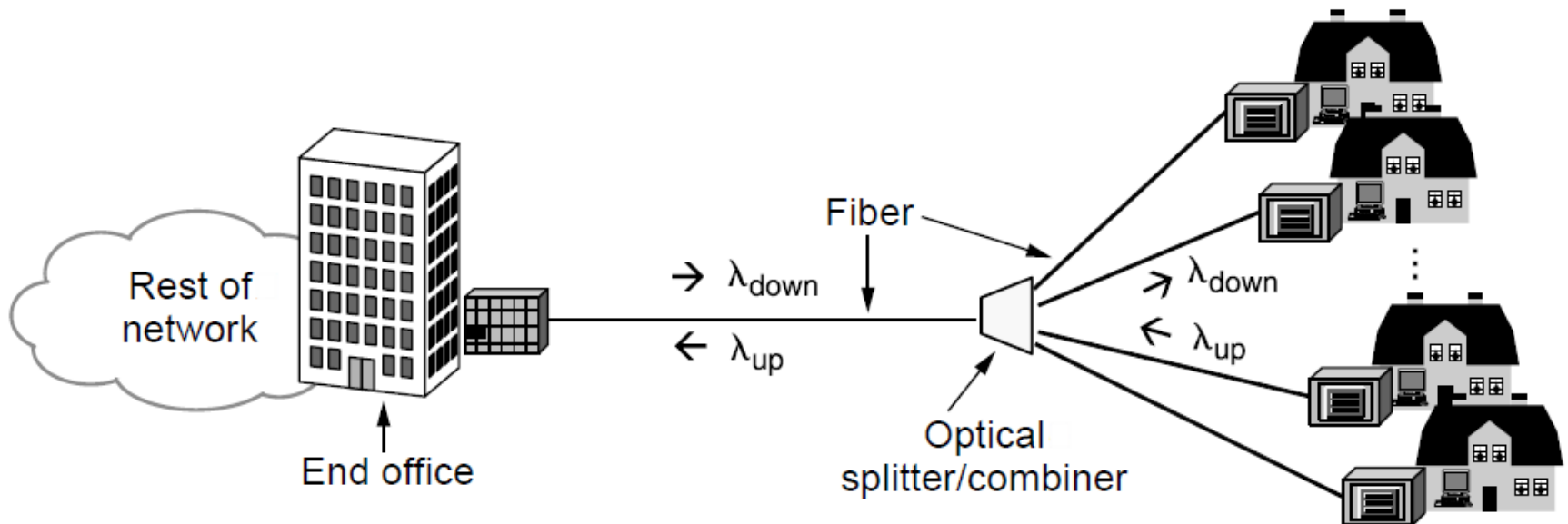
Operation of ADSL using discrete  
multitone modulation.

# Digital Subscriber Lines (3)



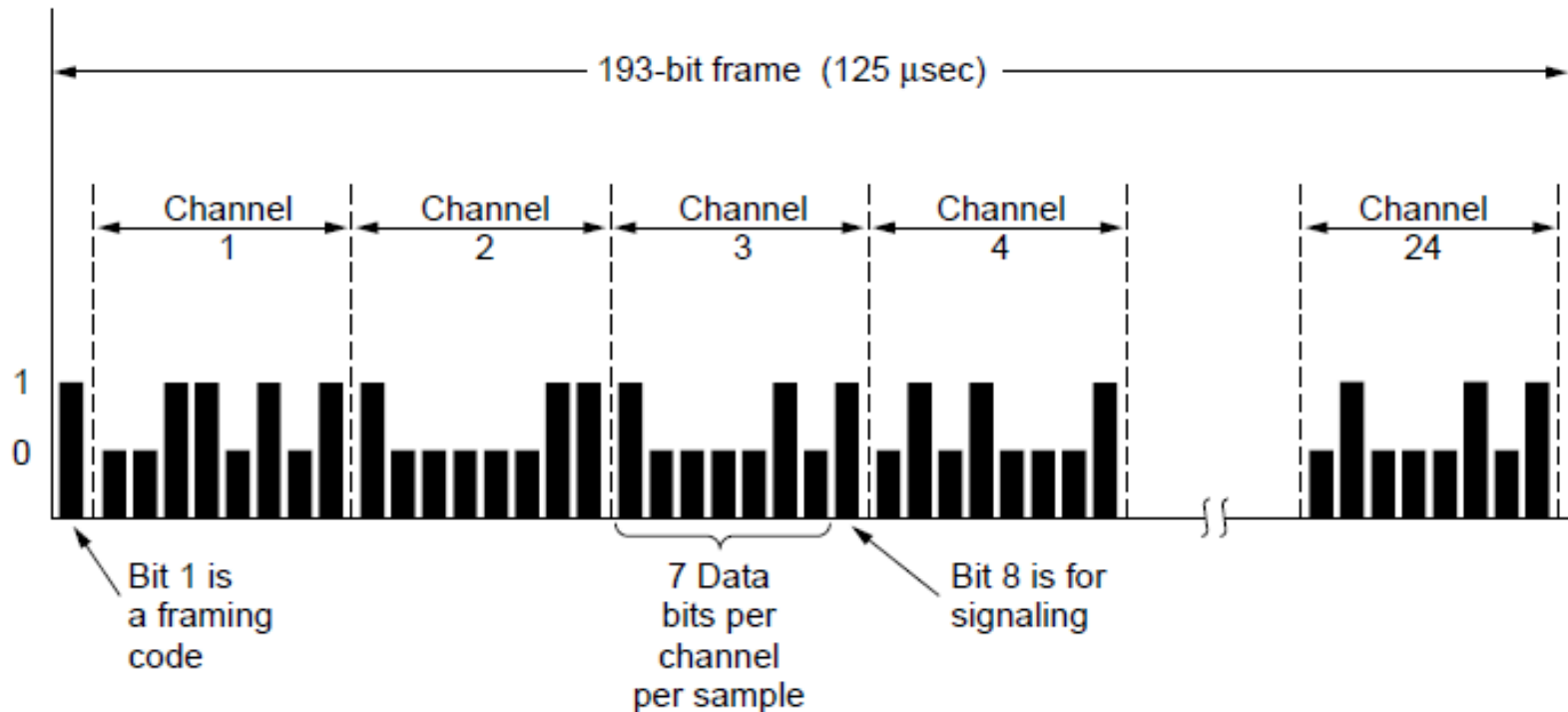
A typical ADSL equipment configuration.

# Fiber To The Home



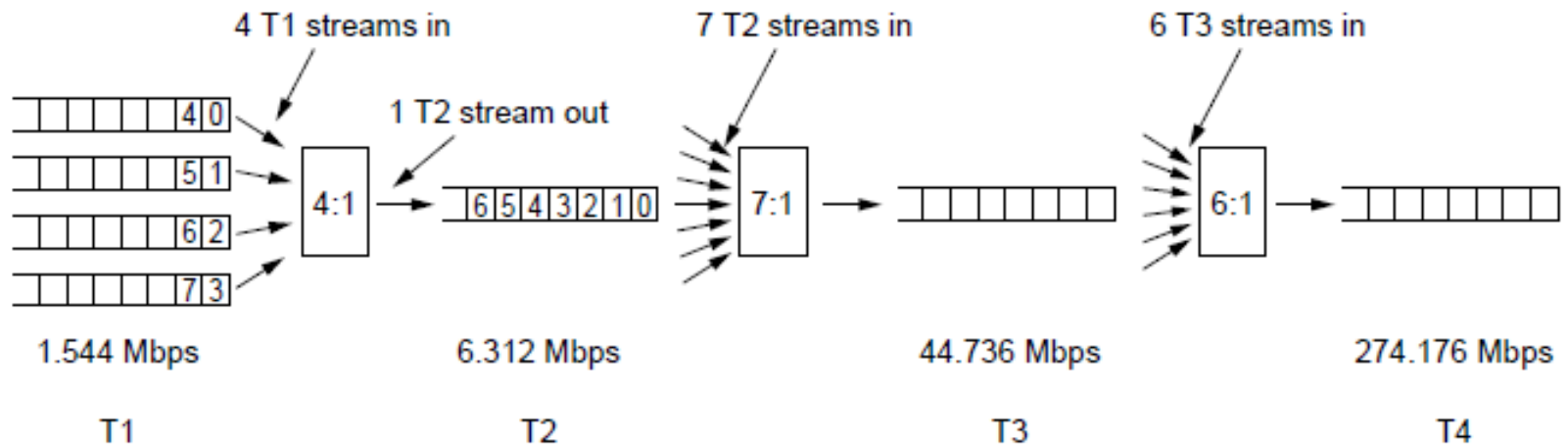
Passive optical network for Fiber To The Home.

# Time Division Multiplexing (1)



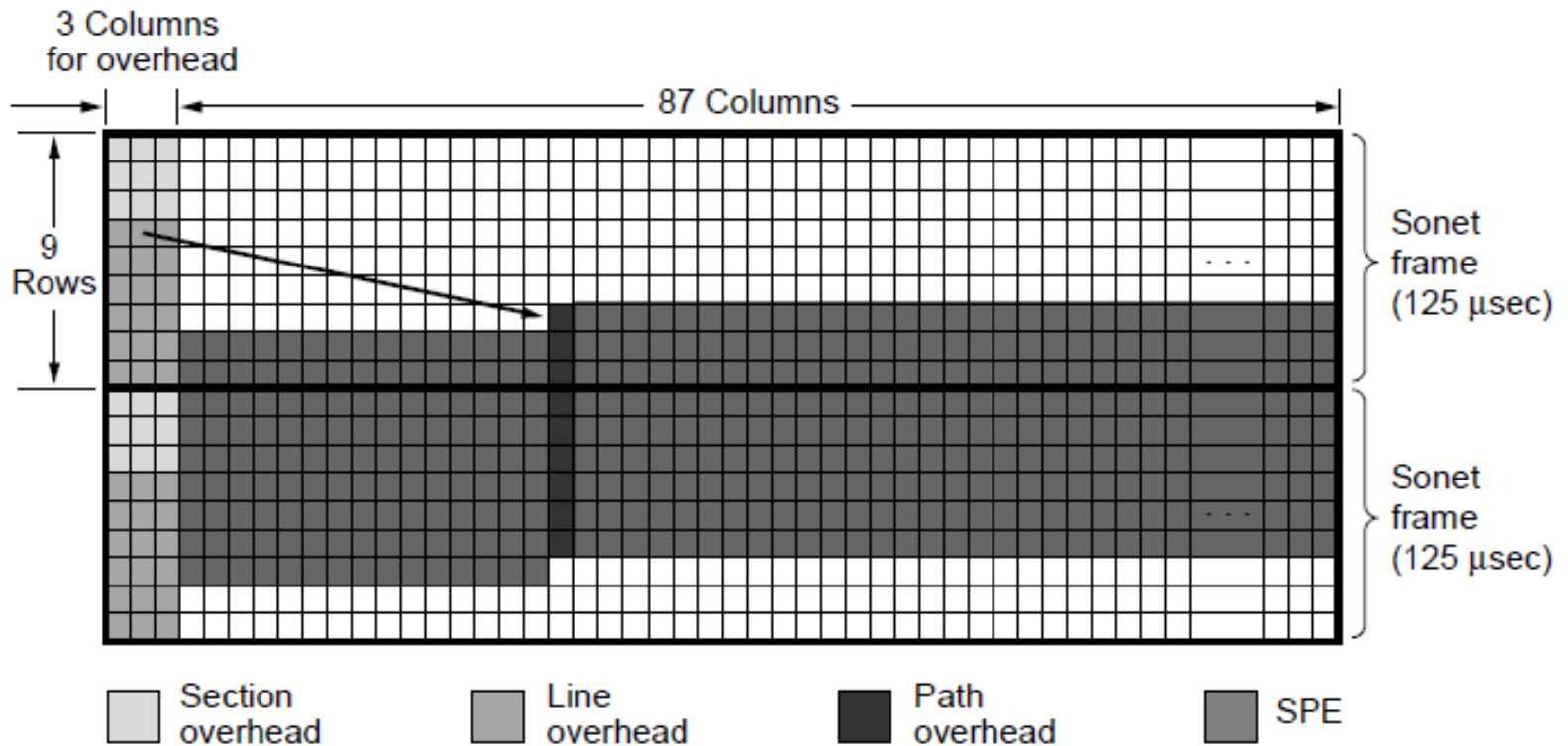
The T1 carrier (1.544 Mbps).

# Time Division Multiplexing (2)



Multiplexing T1 streams into higher carriers

# SONET/SDH (1)



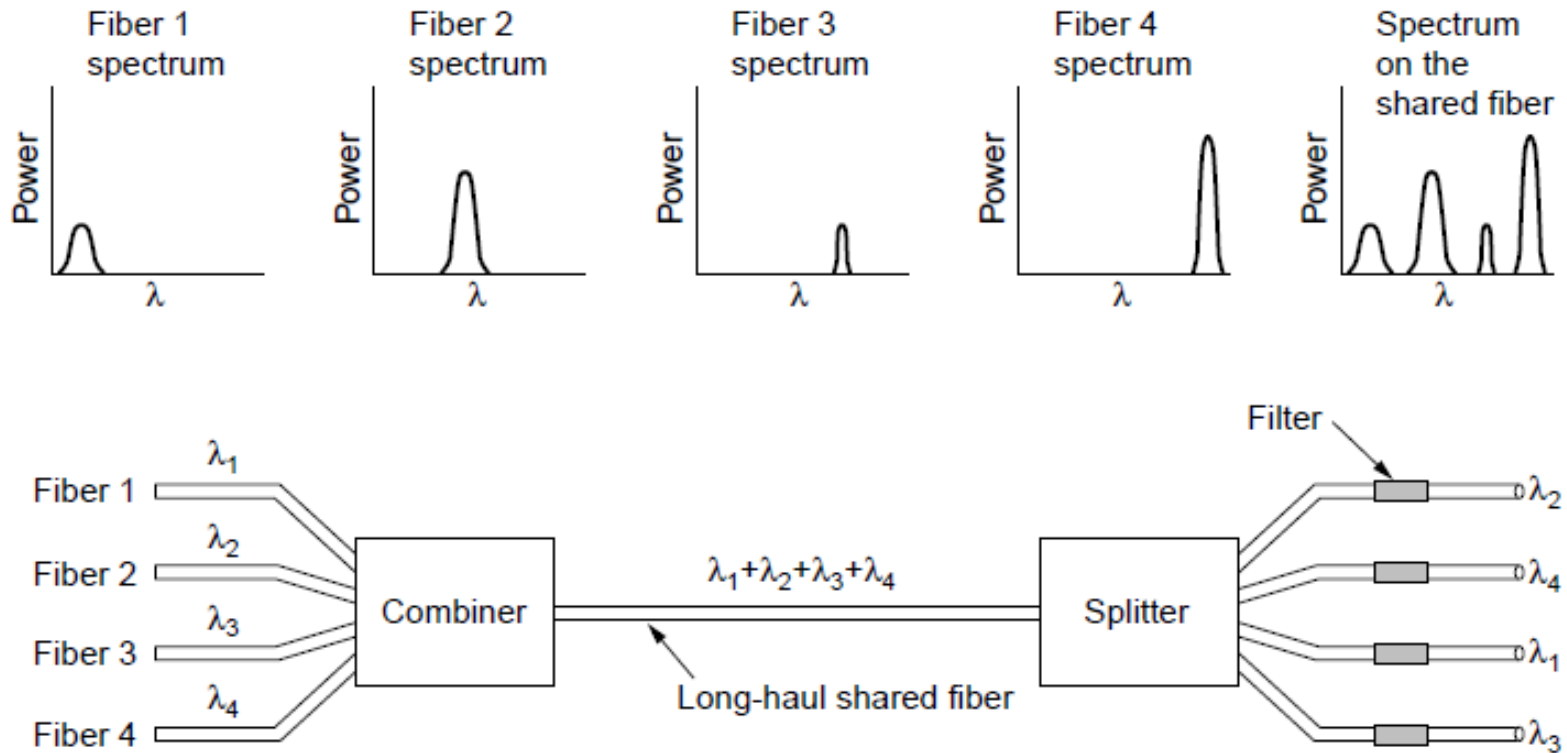
Two back-to-back SONET frames.

# SONET/SDH (2)

| SONET      |         | SDH     | Data rate (Mbps) |           |           |
|------------|---------|---------|------------------|-----------|-----------|
| Electrical | Optical | Optical | Gross            | SPE       | User      |
| STS-1      | OC-1    |         | 51.84            | 50.112    | 49.536    |
| STS-3      | OC-3    | STM-1   | 155.52           | 150.336   | 148.608   |
| STS-12     | OC-12   | STM-4   | 622.08           | 601.344   | 594.432   |
| STS-48     | OC-48   | STM-16  | 2488.32          | 2405.376  | 2377.728  |
| STS-192    | OC-192  | STM-64  | 9953.28          | 9621.504  | 9510.912  |
| STS-768    | OC-768  | STM-256 | 39813.12         | 38486.016 | 38043.648 |

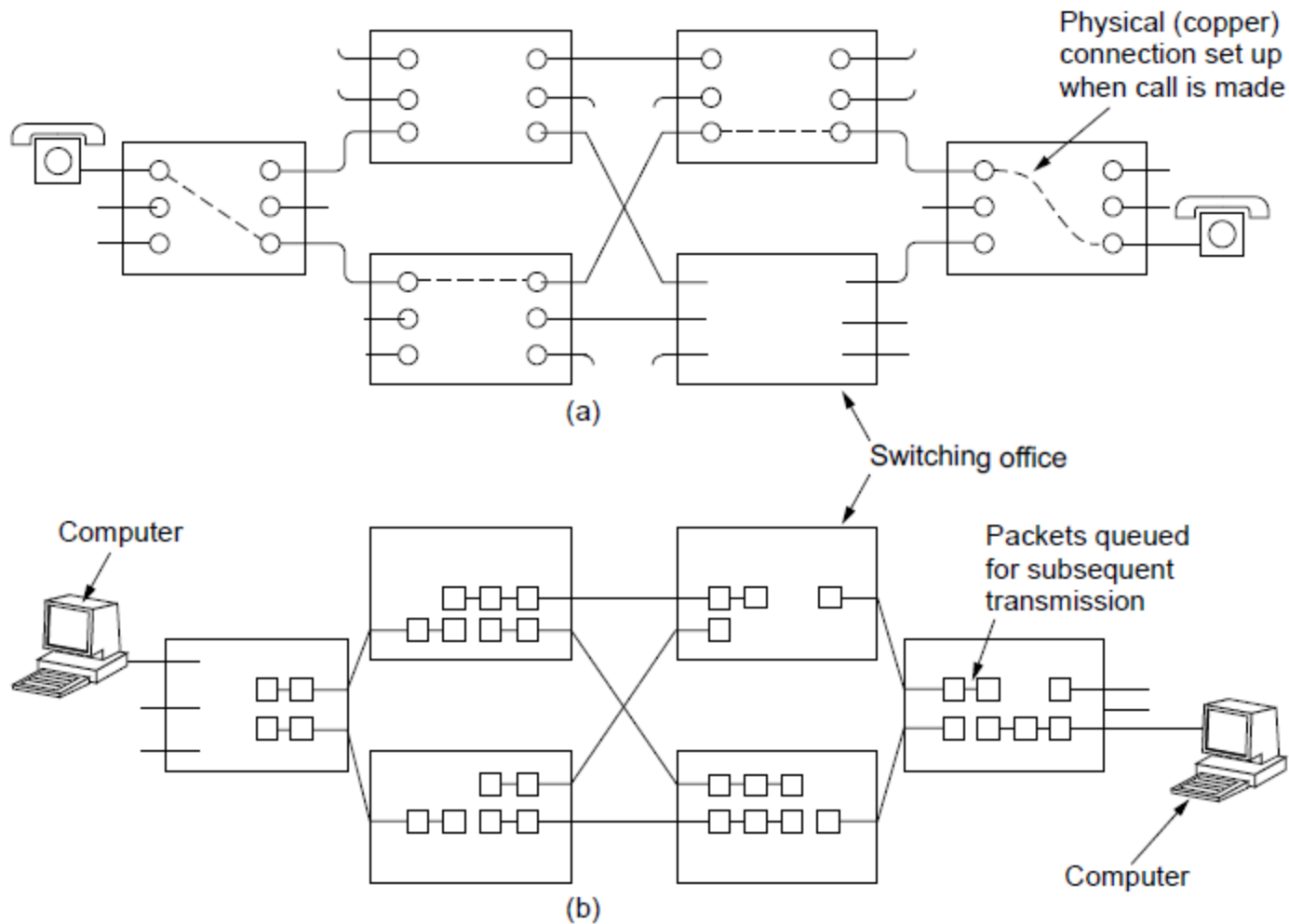
SONET and SDH multiplex rates.

# Wavelength Division Multiplexing



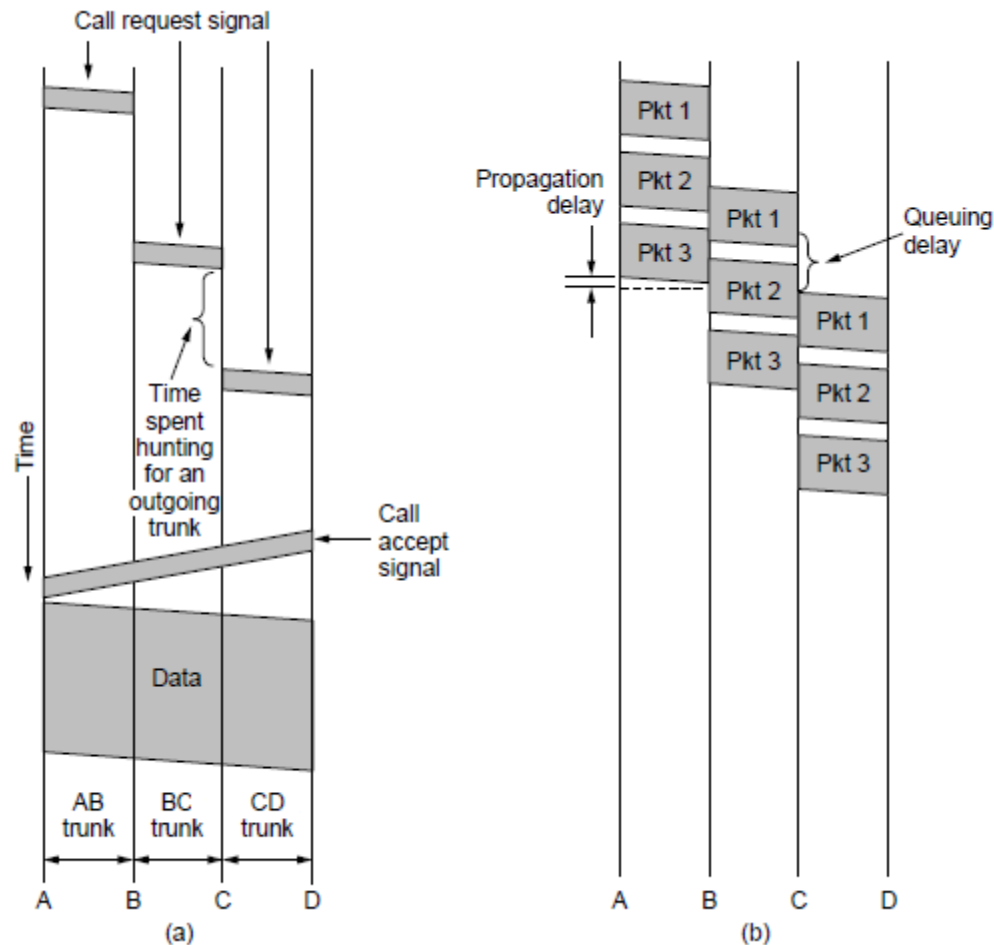
Wavelength division multiplexing

# Circuit Switching/Packet Switching (1)



(a) Circuit switching. (b) Packet switching.

# Circuit Switching/Packet Switching (2)



Timing of events in (a) circuit switching,  
(b) packet switching

# Circuit Switching/Packet Switching (3)

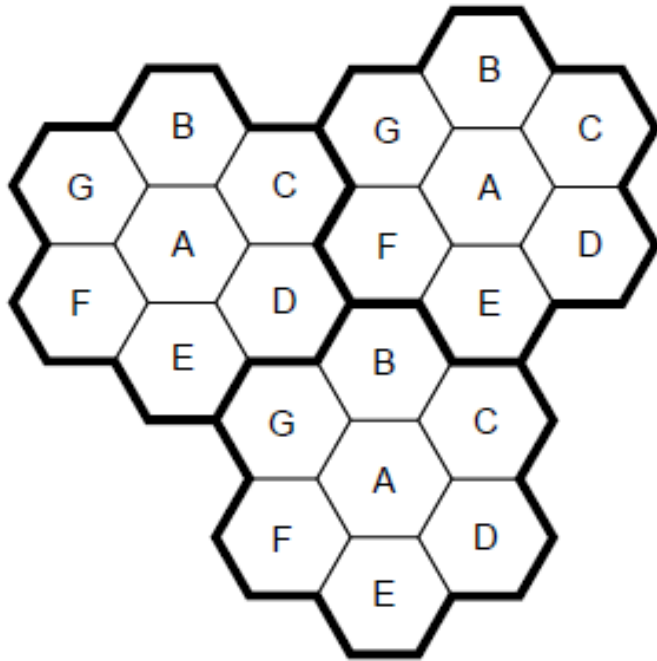
| Item                               | Circuit switched | Packet switched |
|------------------------------------|------------------|-----------------|
| Call setup                         | Required         | Not needed      |
| Dedicated physical path            | Yes              | No              |
| Each packet follows the same route | Yes              | No              |
| Packets arrive in order            | Yes              | No              |
| Is a switch crash fatal            | Yes              | No              |
| Bandwidth available                | Fixed            | Dynamic         |
| Time of possible congestion        | At setup time    | On every packet |
| Potentially wasted bandwidth       | Yes              | No              |
| Store-and-forward transmission     | No               | Yes             |
| Charging                           | Per minute       | Per packet      |

A comparison of circuit-switched and packet-switched networks.

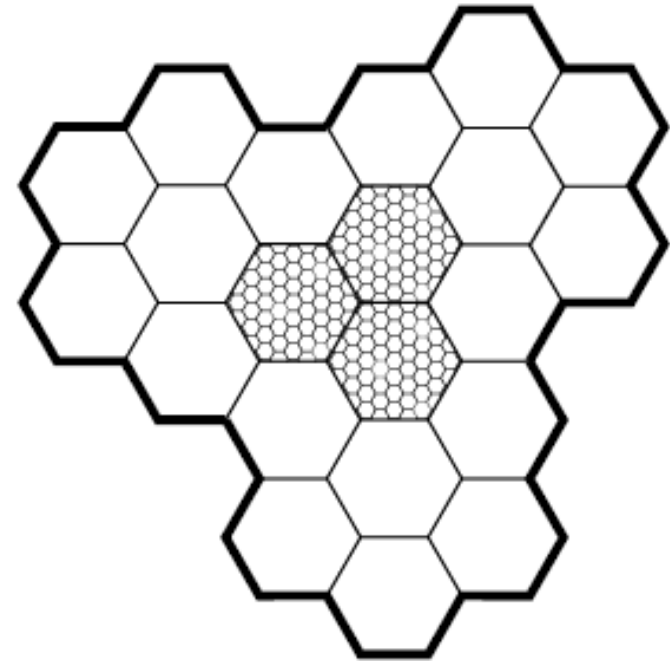
# Mobile Telephone System

- First-Generation (1G) Mobile Phones Analog Voice
- Second-Generation (2G) Mobile Phones Digital Voice
- Third-Generation (3G) Mobile Phones Digital Voice + Data

# Advanced Mobile Phone System



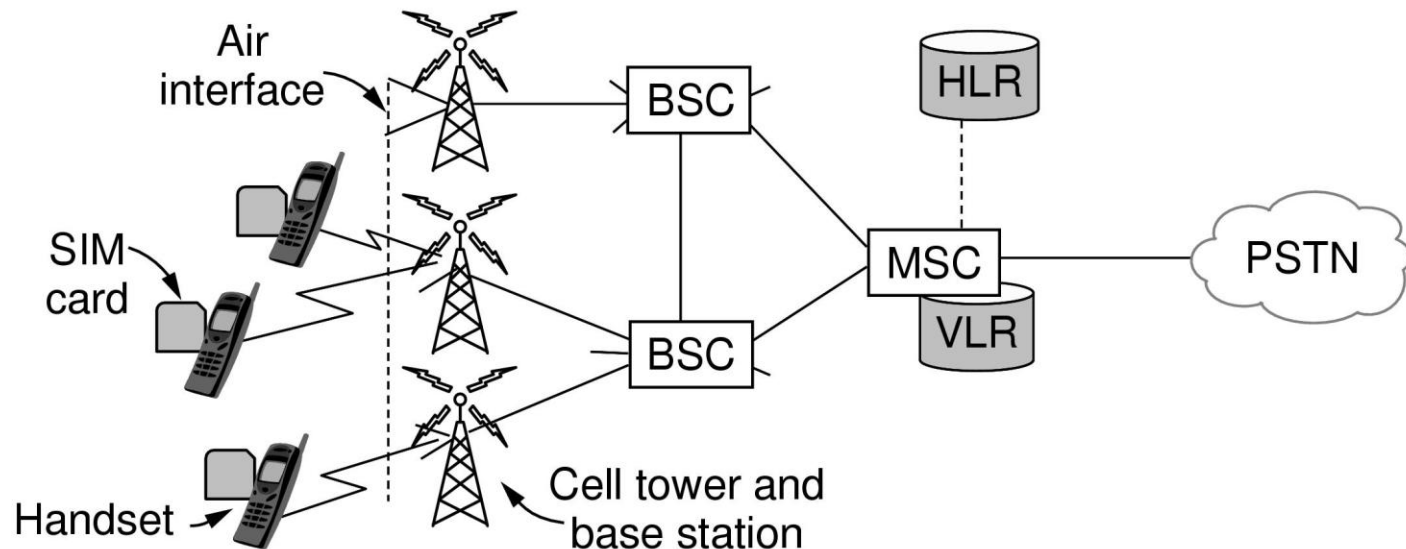
(a)



(b)

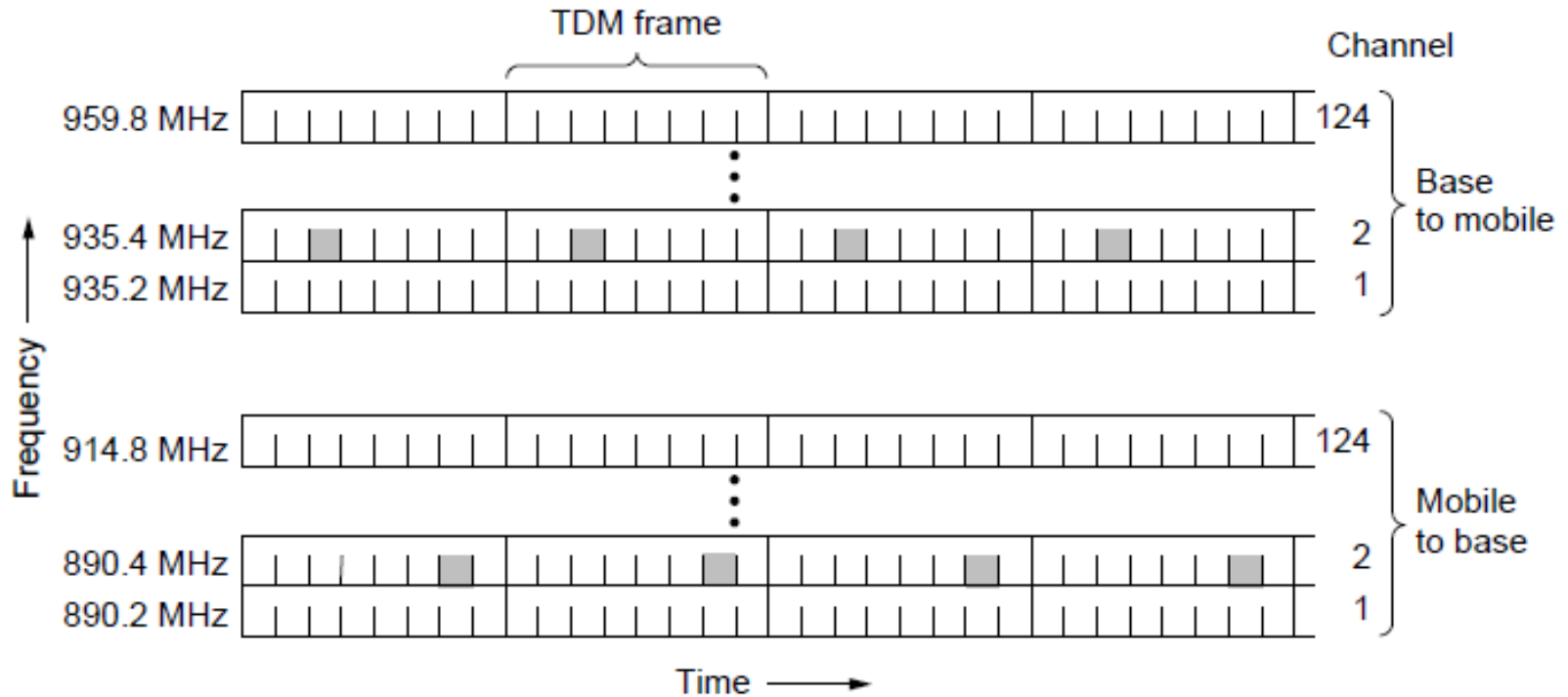
- (a) Frequencies are not reused in adjacent cells.
- (b) To add more users, smaller cells can be used.

# GSM—The Global System for Mobile Communications (1)



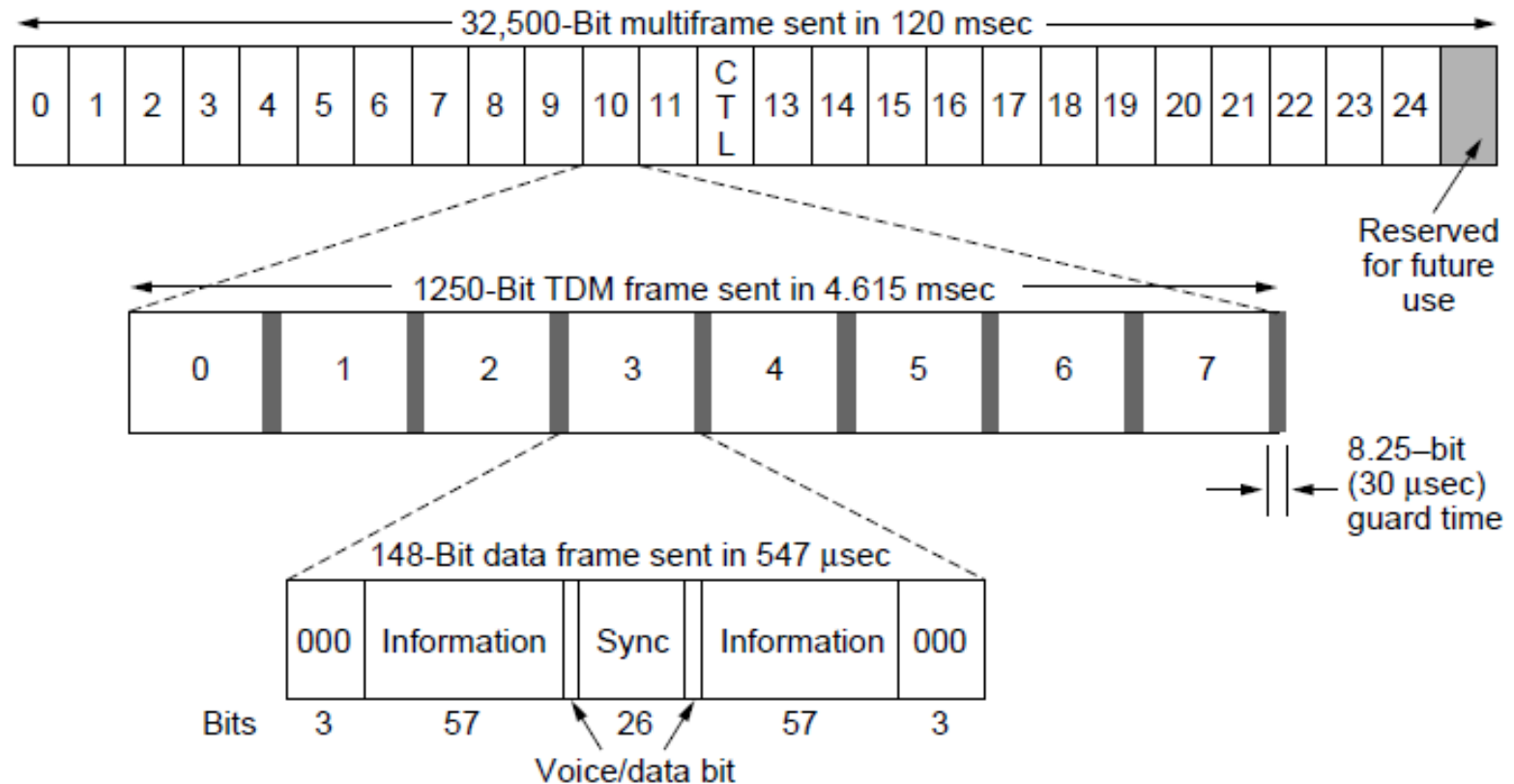
GSM mobile network architecture.

# GSM—The Global System for Mobile Communications (2)



GSM uses 124 frequency channels, each of which uses an eight-slot TDM system.

# GSM—The Global System for Mobile Communications (3)



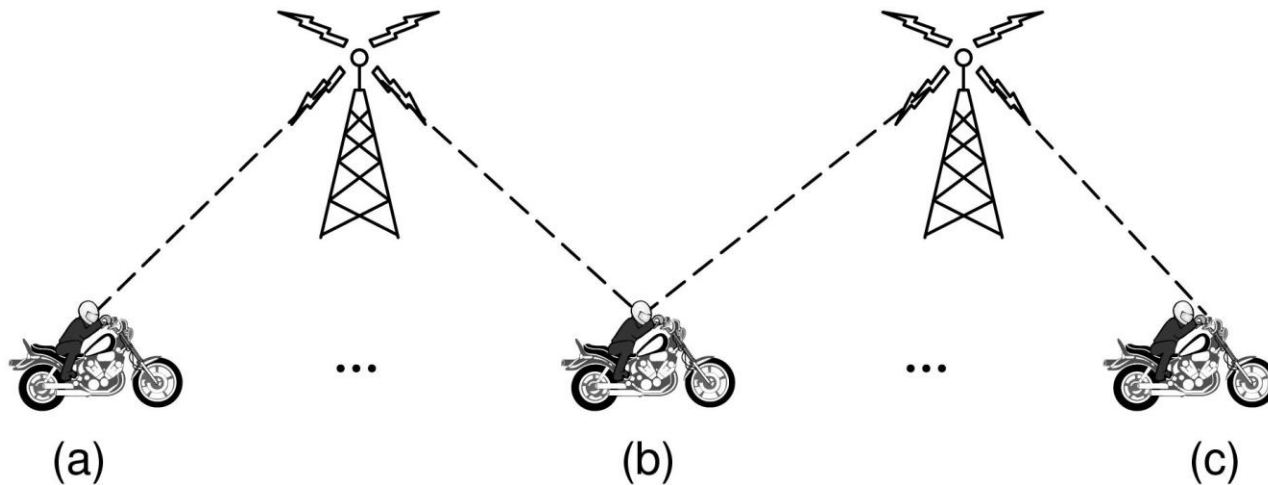
A portion of the GSM framing structure.

# Digital Voice and Data (1)

Basic services intend by IMT-2000 network

- High-quality voice transmission.
- Messaging (replacing email, fax, SMS, chat).
- Multimedia (music, videos, films, television).
- Internet access (Web surfing, incl. audio, video).

# Digital Voice and Data (2)

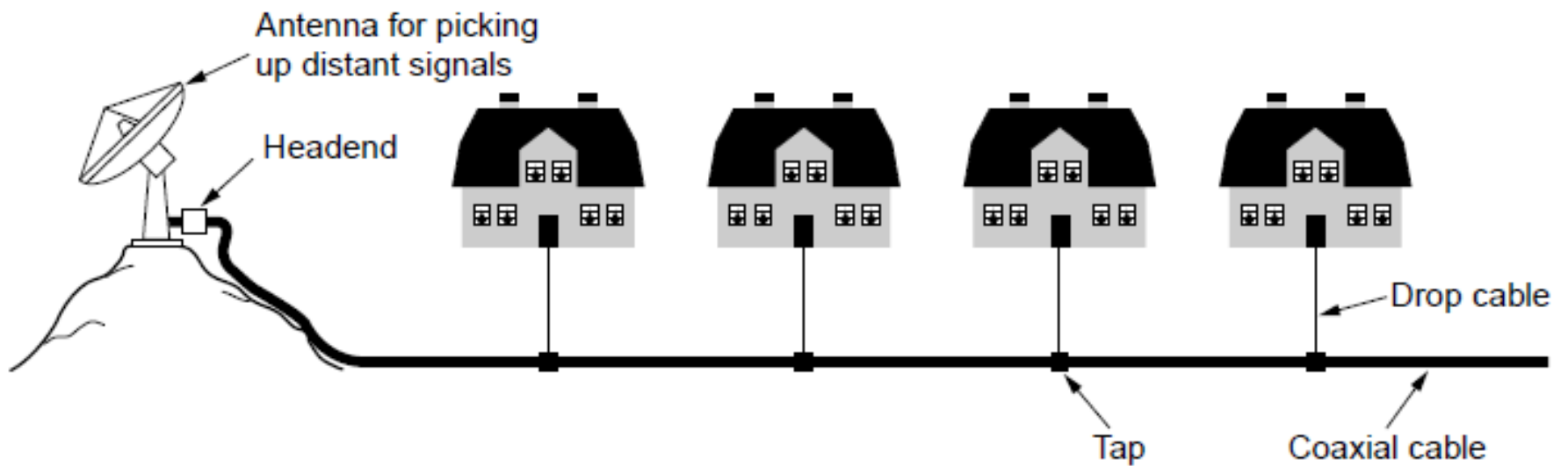


Soft handoff (a) before, (b) during, and (c) after.

# Cable Television

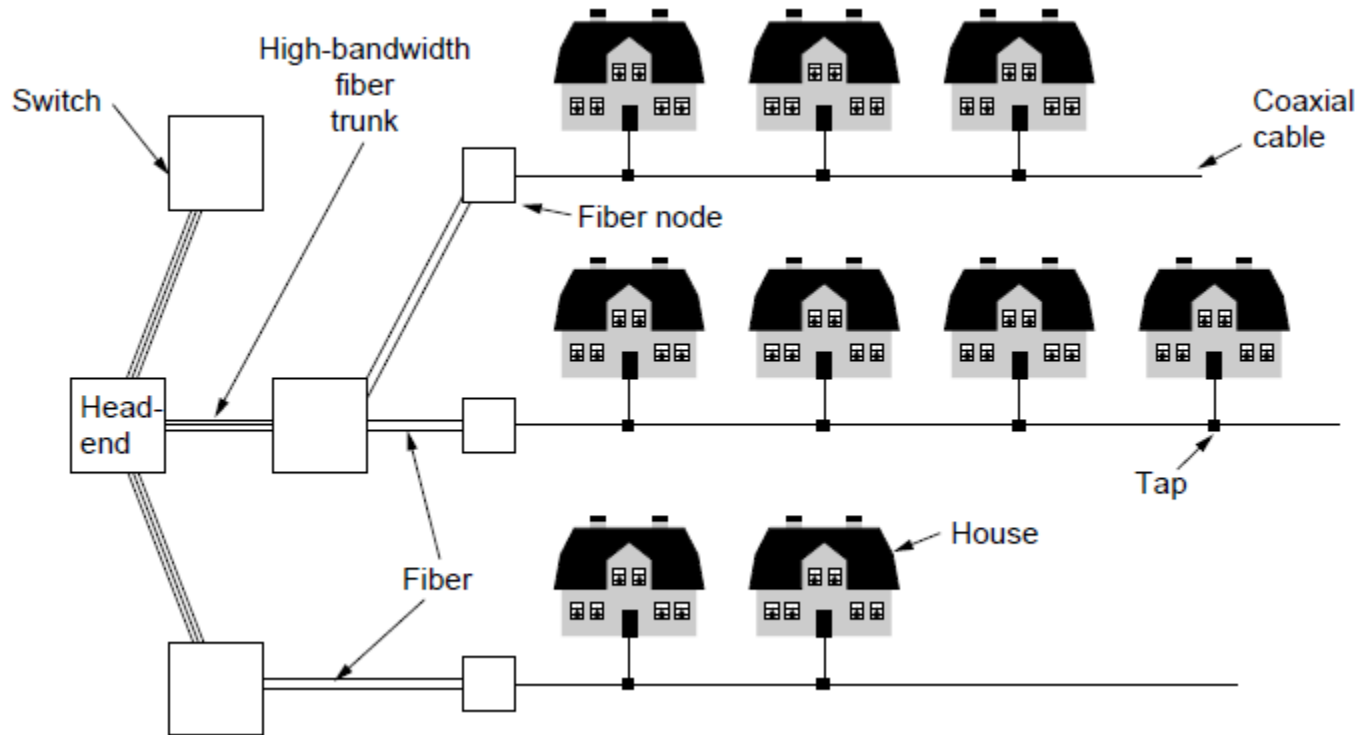
- Community antenna television
- Internet over cable
- Spectrum allocation
- Cable modems
- ADSL versus cable

# Community Antenna Television



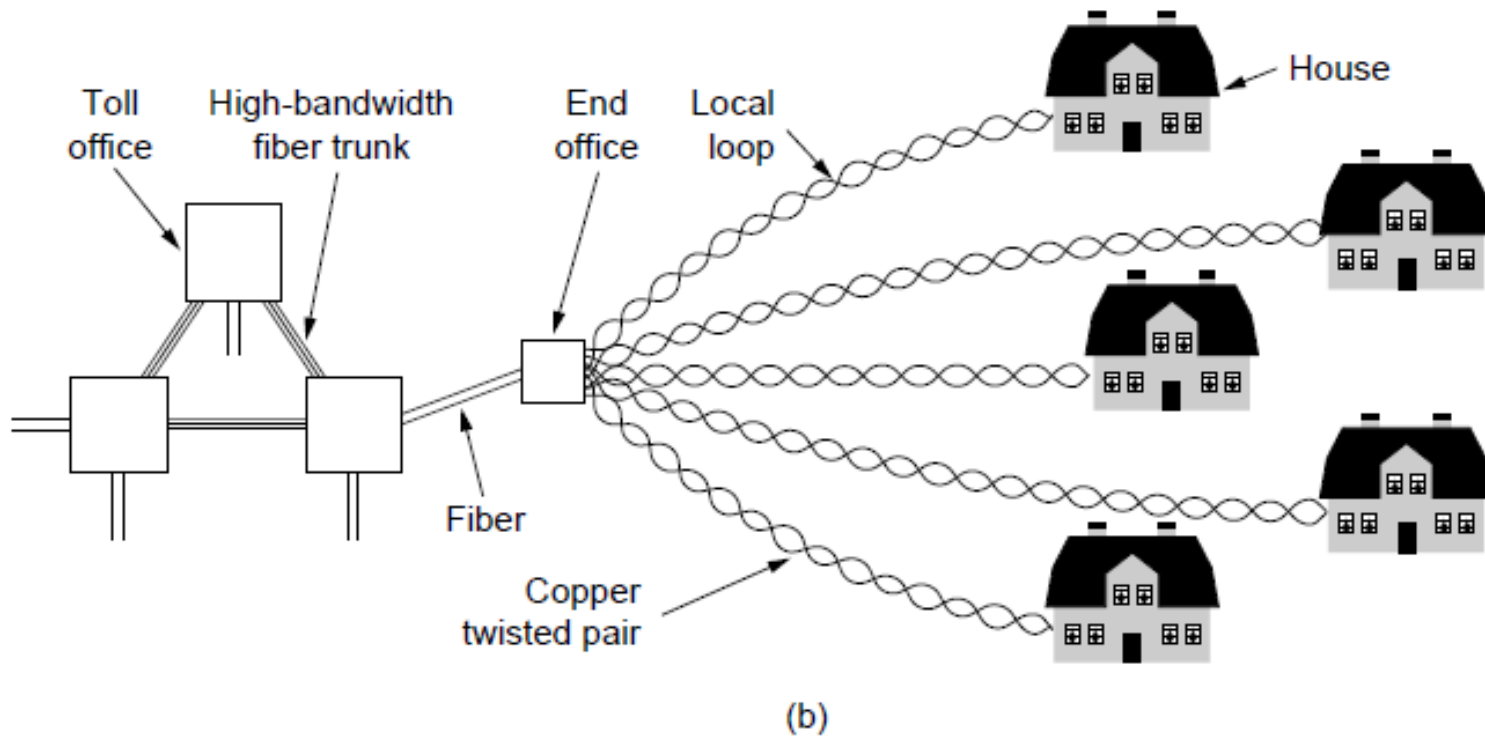
An early cable television system

# Internet over Cable (1)



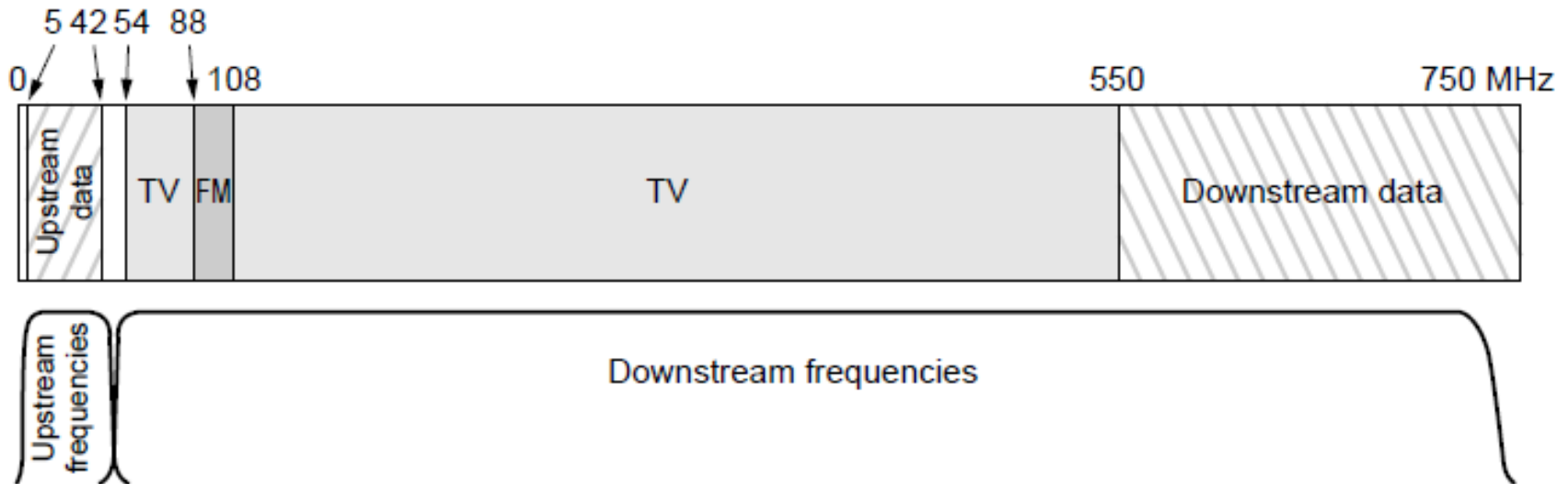
Cable television

# Internet over Cable (2)



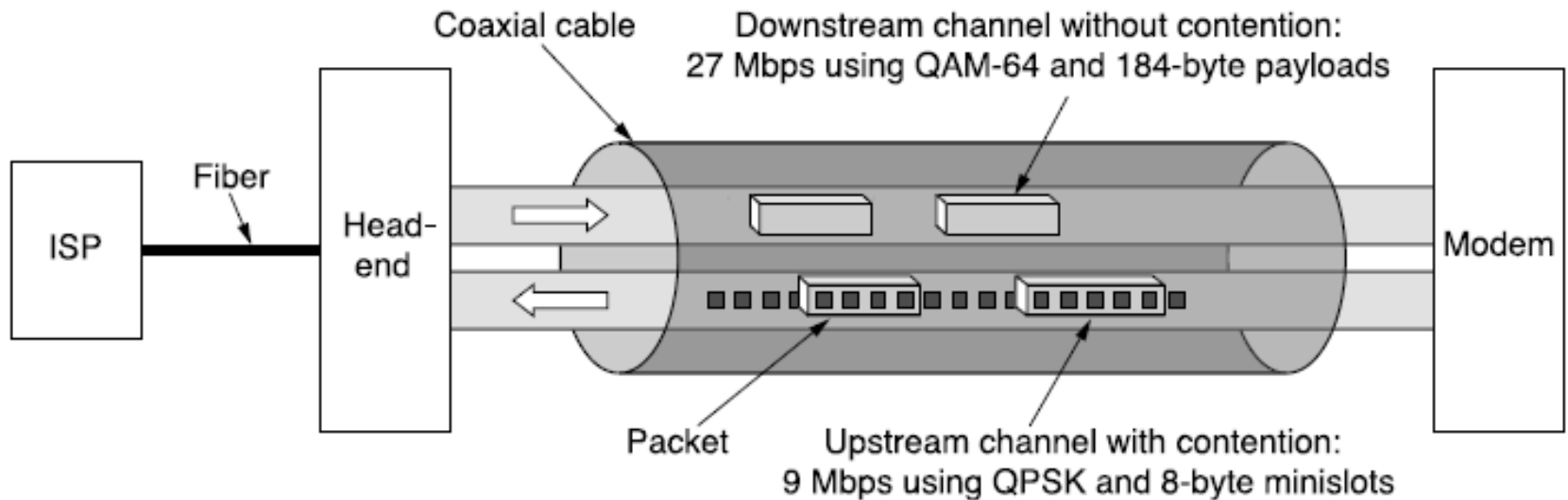
The fixed telephone system.

# Spectrum Allocation



Frequency allocation in a typical cable TV system used for Internet access.

# Cable Modems



Typical details of the upstream and downstream channels in North America.

End

Chapter 2